

DODOSIM

206 V2



Installation and Configuration

Applicable to DodoSim 206 V2 for:

**Microsoft Flight Simulator X,
Lockheed Martin Prepar3D**

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1 INTRODUCTION

This document provides instructions for installing and configuring the DodoSim 206 V2 add-on aircraft in both Prepar3D and Microsoft Flight Simulator X.

It is applicable to all versions of the add-on, specifically:

- DodoSim 206 V2 P3D – Home (non-commercial use) version
- DodoSim 206 V2 FSX – Home (non-commercial use) version

The DodoSim 206 V2 is available in both 32 and 64 bit versions. The 64 bit version is only applicable to Prepar3D v4 onwards. Ensure you have the correct version available for your type of use and platform. The Home versions are not licenced to end users for commercial gain.

Where specifics between the various versions differ, this shall be noted accordingly.

2 DODOSIM 206 V2 INSTALLATION

Before installing, ensure any previous version is removed. If installed from an installer, run the uninstall option from Windows Control Panel. If changes have been made since installation, such as additional of any paint kits, etc. these will require manual deletion. Ensure any DodoSim 206 V2 files are removed from the simulation's Gauges, Effects and SimObjects folders. Specifically, these files are:

```
{application root}\gauges\DodoSim206.gau  
{ application root }\gauges\Dodosim206HiDef.CAB  
{ application root }\SimObjects\Rotorcraft\DodoSim_206_{all variants}
```

The *{ application root }* directory is the root of the installation for the simulation platform, whether it be for Microsoft Flight Simulator X, Microsoft Flight Simulator X Steam Edition, Lockheed Martin Prepar3D V3 or V4, i.e.:

```
C:\Program Files (x86)\Microsoft Games\Microsoft Flight Simulator X  
C:\Program Files (x86)\Steam\steamapps\common\FSX  
C:\Program Files (x86)\Lockheed Martin\Prepar3D v3  
C:\Program Files\Lockheed Martin\Prepar3D v? (where ? is 4 upwards)
```

Take care not to remove any non-DodoSim 206 V2 files. The installer is unable to overwrite pre-existing files and so failure to clean out a previous version may result in an incorrect new installation.

To install the DodoSim 206 V2, locate and run the respective setup.exe file. When prompted to do so, verify the application install path. Due to the various versions of Prepar3D and Microsoft Flight Simulator X now available, the installer makes no attempt to locate the installation directory, instead simply suggesting the default install path. Change this path if required to point to the location of FSX.exe or Prepared3D.exe as required and described above.

3 PREPAR3D/FLIGHT SIMULATOR X SETTINGS

Lockheed Martin Prepar3D and Microsoft flight Simulator X provide the same settings through the Options->Settings menu dialog. Specific locations of the settings may differ between the applications. Illustration is given for the Prepar3D V3 dialogs.

Prepar3D and Microsoft Flight Simulator X are henceforth referred to within this document as “the platform” unless differences necessitate they be referred to specifically.

The platform’s settings are accessed via the program menu thus:

Note: Screenshots are taken from Prepar3d v3. FSX and later P3D versions may differ in navigability and layout, but provide the same functionality.



Figure 1 - Simulation settings menu

3.1 SETTINGS – GENERAL

General settings should be set up as per the image below, unless specific training installation requires otherwise:

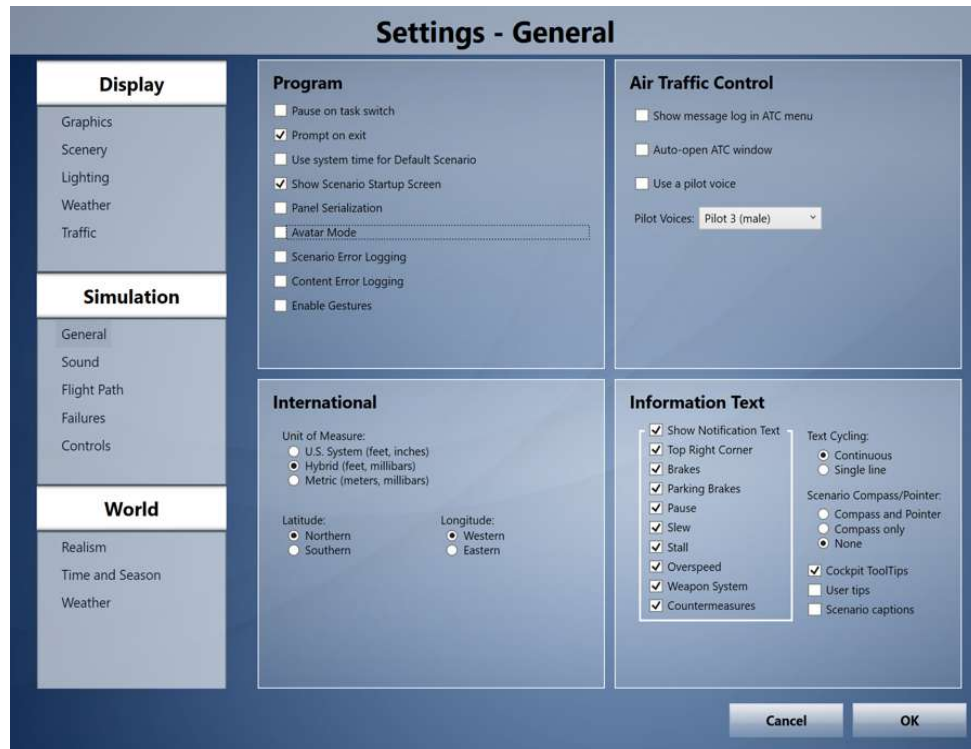


Figure 2 - Simulation General Settings

De-selecting “Pause on task switch” ensures the simulation continues running if Windows application focus is changed by the user to another program.

3.2 SETTINGS – SOUND

The DodoSim 206 V2 includes a DirectSound sound system that works alongside the platform's own. However, since the various volume level settings used by the platform are not exposed through the platform's Software Developer Kit (SDK) Application Programmer's Interface (API), additional slider controls are required to balance the DodoSim 206 V2's sound levels to those of the platform, (detailed later in this document.)

Initially, set the platform's sound levels to default levels, as below. If balancing is required later between platform's sound levels (used by ATC, other aircraft and the environment) and the DodoSim 206 V2 (engine and rotor sounds, warning sounders, etc.), adjustments can be made to either the platform's sound levels or the DodoSim 206 V2 as appropriate.



Figure 3 - Simulation Sound Settings

3.3 SETTINGS – CONTROLS

The Controls settings section is used to map input device axes to simulator functions and sets their sensitivities.

Assignment is performed in the Axes tab and are assigned by double-clicking on the axis name and then moving the required axis on the controller, (whilst carefully making sure no other axis is moved at the same time to ensure the wrong axis is not detected and assigned.)

Controller axis to simulation events should be configured as such:

Simulation Event	Controller Axis
Aileron axis	Cyclic left/right
Elevator axis	Cyclic fore/aft
Propeller axis	Engine throttle
Throttle axis	Collective pitch
Rudder axis	Anti-torque pedals
Flaps axis	Rotor brake (if desired)

Table 1- Controller axis mappings

Note that it may transpire that some axes require reversing in the settings, dependent on how the controller is wired. This reversal can either be applied here in the Settings dialog, in which case it will apply to all aircraft within the simulation, or it can be applied only to the DodoSim 206 V2 specifically from within the DodoSim 206 V2's own Settings dialog, (described later within this document.)

Once assigned, the axes should be set to maximum sensitivity and minimum deadzone on the Calibration tab, i.e.:

**Figure 4 - Simulation Controls Settings**

Prepar3D provides an additional facility over FSX to permit the axis ranges to be measured such that axis extents and centre points can be accurately calibrated:

3.4 SETTINGS – REALISM

Realism settings should be set up as per the image below, unless specific training installation requires otherwise:



Figure 5 - Simulation Realism Settings

All flight model settings should be set to most realistic to ensure that all the DodoSim 206 V2's flight characteristics behave as designed and expected.

Opting to ignore crashes simplifies the training environment, since an overly hard landing will cause the simulation scenario to reset.

Ensure that the options to ignore weights and forces are not enabled, otherwise aircraft fuel and load-out do not appropriately impact handling and performance.

4 DODOSIM 206 V2 SETTINGS

The DodoSim 206 V2 specific settings can be accessed when the aircraft is loaded in the simulation using the following menu:

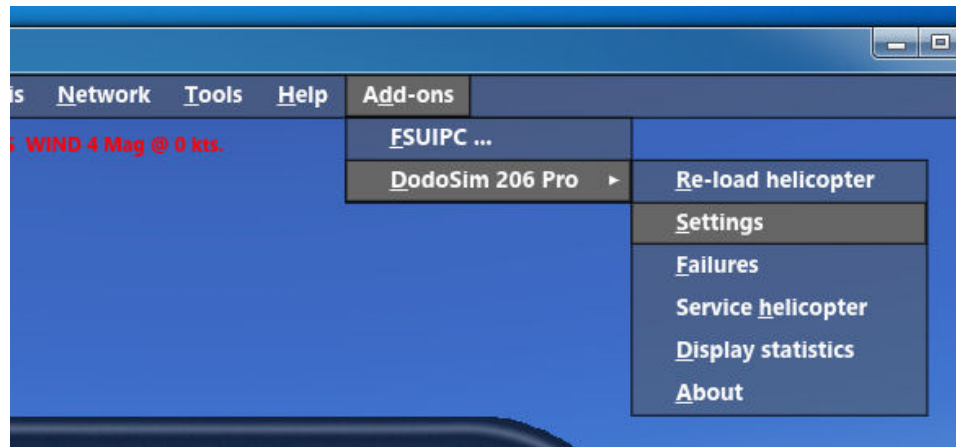


Figure 6 – DodoSim 206 V2 Settings menu option

(Note that this menu is only available when the DodoSim 206 V2 is loaded as the user's aircraft in the simulation.)

Selecting this menu will cause the simulation to suspend and open the DodoSim 206 V2 Settings dialog window. The Settings dialog has three tab pages, splitting settings into three functional groups General settings, Controls settings and Sounds settings. The following subsections describe each of these tab pages in detail.

4.1 GENERAL SETTINGS

DodoSim 206 V2 P3D - Settings

dodo **DodoSim 206 V2 P3D** Settings

General | Controls | Sounds | Dynamics

☒ Use costings ☒ Save persistent aircraft data ?

☐ Rotor only Virtual Cockpit None Force show main panel ?

☒ Auto open/close panels during auto start/stop sequence

Systems Failure Mode ?

☐ Cumulative wear ☒ Timers ?

☒ Display failure hints ☒ Failure pops CB

☐ CB reset clears failure

Equipment ?

☒ SAS fitted by default ☒ SAS used by default

☐ Battery instant recharge

SAS / Autopilot Set-Up ?

15 Bypass cyclic threshold % 0.8 Deviation time-out (secs) ?

From Centre Deviation detection method

Proportional disconnect Bypass disconnect method

Chime + Flashing G Bypass indication Flashing Red Saturated indication

60 Max cyclic pitch authority % ☒ Play Nav capture chime

60 Max cyclic bank authority % ☐ External actuators

☐ Disable limiters

Flight1 GPS ☒ Trim hat bump extended functions

☐ CTS is relative ☐ Link CRS

Instrumentation ?

☐ Simple AI gyro ☐ AI self-erects 95 AI rigidity 1200 VFR squawk code ?

RCC ADF default type

Standard Default Annunciator Layout

This Aircraft (G-DODO) ?

Bendix Fuel Control Unit style Reset persistent data for G-DODO ?

RCC ADF type ☐ ADF Flight timer ground sense

☒ Has SAS fitted Alternative 1 Annunciator Layout

Close

Figure 7 – DodoSim 206 V2 General settings

Note that each section within any of the settings groups is accompanied by a “?” icon. Clicking this will open a dialog briefly describing the purpose of each setting, in supplement to this manual, i.e.

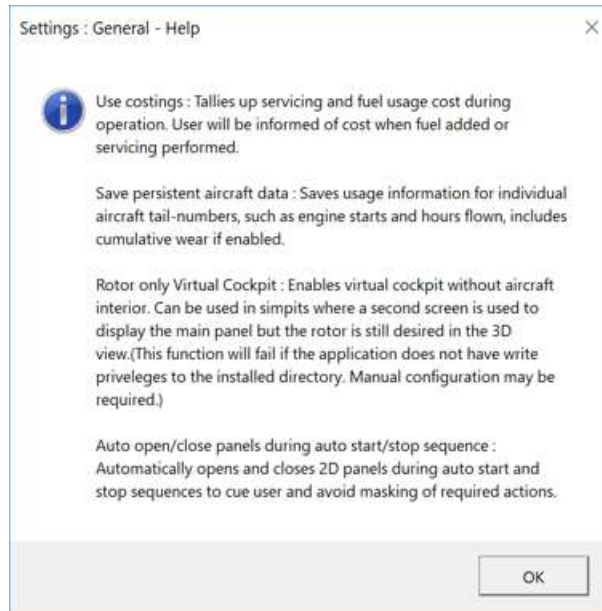


Figure 8 – DodoSim 206 V2 Settings hints

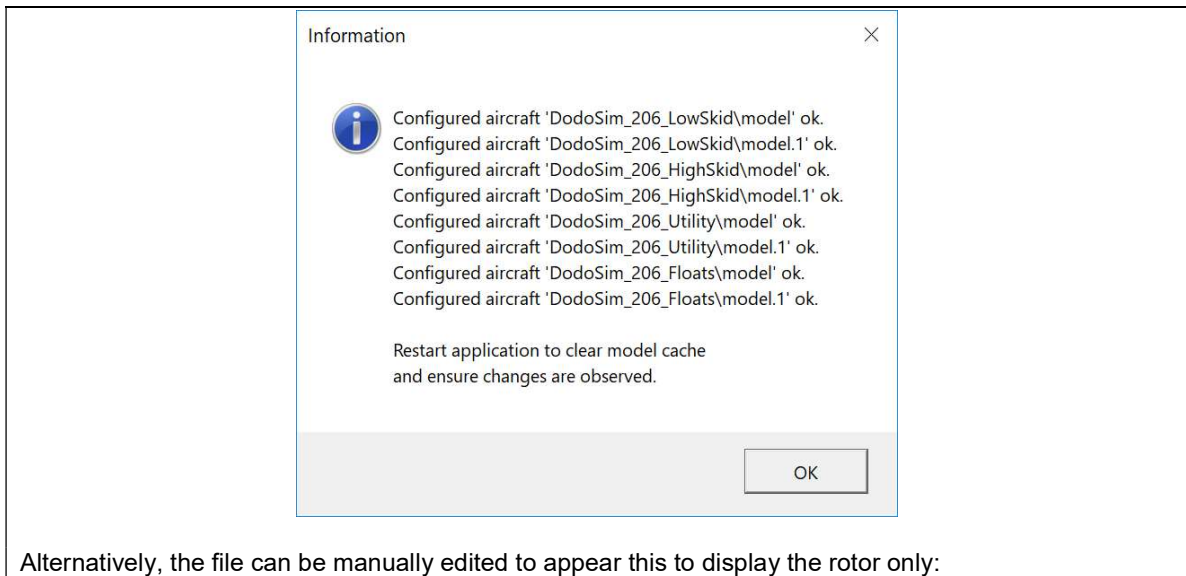
The General Settings dialog tab options are split into five groups:

1. **General Preferences** - Applies global options to:
 - a. **Use costings** - Uses hypothetical costing rates for aircraft fuel purchases and servicing in “Cumulative wear” Failure Mode.
 - b. **Save persistent aircraft data** - Maintain persistent damage, start-up counts and flight hours in “Cumulative Wear” Failure Mode.
 - c. **Rotor only Virtual cockpit** – Setting this option causes the object 3D model description file to be replaced with one that includes only the main rotor, (removing the entire cabin model), when the simulation is in virtual cockpit (VC) mode. Un-checking this reverts the configuration to the default rotor and cabin model. This function is most applicable to hardware cockpit set-ups, enabling the user to display the 2D main panel on a separate monitor whilst still being able to observe the rotor in the main out the front canopy when running in virtual cockpit mode.

Note: In order for the program to modify the configuration file (located in `{application root}\SimObjects\Rotorcraft\DodoSim_206_{variant}\model\model.cfg`), the application must have administrative privileges to read and write to within the `Program Files` directory. You may need to run the simulation in admin mode (right-click the program's shortcut and select 'Run as administrator'), or give the user account administrative rights over the `{application root}` directory for this settings function to work.

When checked/un-checked, the success of the configuration files modifications will be output, for example:

DodoSim 206 V2 Installation and Configuration



```
[models]
normal=DodoSim206
interior=..\..\DodoSim_206_LowSkid\model\DodoSim206_interior_noVC.MDL
```

Or this, to restore the full virtual cockpit:

```
[models]
normal=DodoSim206
interior=DodoSim206_interior
```

- d. **Auto open/close panels during auto start/stop sequence** - When checked, this option causes the aircraft to open the applicable 2D panel window when a particular action is required to be taken during a cued start-up or shut-down sequence, (such as flicking a switch.) When unchecked, it is the responsibility of the pilot to ensure the correct panel is open when being prompted for an action. (If the specific panel is not open or obscured, the pilot may not be able to see the cue prompt.)

Note: When activated, trainer mode steps the pilot through the aircraft start-up and shut-down procedures and is detailed more thoroughly in the DodoSim 206 V2 User Manual.

2. **Systems Failure Mode** – Sets global options for the failure modes:

- a. **Cumulative wear/Timers** – Sets the failure mode to either:
- Cumulative Wear; whereby rough handling of the aircraft causes damage over time resulting in eventual system failure at an unpredictable point in time,
 - Timers; whereby the DodoSim 206 V2 Failures menu option can be accessed and utilised in much the same way as the default platform Failures dialog to specify a point in time at which the specified system will fail.
- b. **Display failure hints** – When checked, the pilot will be informed of system failures and accrued damage when they occur via the platform's information bar, i.e.



Figure 9 – DodoSim 206 V2 Failure hint

- c. **Failure pops CB** – Some aircraft electrical systems have a circuit breaker (CB) attached to them. When checked, if a failure occurs in these systems the CB will be “popped”. (The pilot may then opt to try and restore correct operation by attempting to reset the system by pushing the CB back in to energise the circuit.
 - d. **CB reset clears failure** – When checked, a failure in an electrical system that popped a circuit breaker can be cleared and the system restored to normal operation by re-application of the relevant circuit breaker. Note: this only applies to failures instigated via the DodoSim 206 V2 Failures dialog, not failures instigated by the simulation’s default Failures dialog.
3. **Equipment** – applies global settings regarding fitted equipment:
 - a. **SAS fitted by default** – Determines whether a Stability Augmentation System (SAS – including autopilot functionality) is fitted by default to any DodoSim 206 V2 aircraft when a new tail number is set.
 - b. **SAS used by default** – Determines whether the SAS activation is to be included in the trainer mode cued start-up procedure. If set, the pilot will be instructed to energise the SAS using the circuit breakers and activate the SAS master stability mode during the cued start-up procedure.
 - c. **Battery instant recharge** – When checked, the aircraft’s battery will be instantly recharged to full capacity when the engine is running and the generator engaged.
4. **SAS / Autopilot Set-Up** – Applies globally to all SAS installations:
 - a. **Bypass cyclic threshold %** - Determines the percentage of cyclic input deflection required before the SAS temporarily disengages (termed “bypass”), permitting the pilot complete cyclic authority. The SAS will slowly re-engage when the cyclic is returned to within this limit.
 - b. **Deviation time-out (secs)** – The time in seconds that it takes for the SAS to regain complete authority after the cyclic input deflection has been returned to within the threshold percentage.

- c. **Deviation detection method** – This is the method by which the SAS determines that the pilot has moved the cyclic stick and the SAS should temporarily disengage, entering Bypass Mode. For self-centring (spring-loaded) cyclic sticks, it should be set to 'Centre'. For non-self-centring sticks that remain in position when released, it should be set to 'Movement'. If the intention is to not utilise Bypass Mode and instead always fly with the SAS in full authority (called "Flythrough Mode"), this should be set to None.
- d. **Bypass disconnect method** – This determines whether the SAS should reduce its strength when in Bypass Mode proportionally according to how far the pilot has moved the stick from the set position, or fully, relinquishing full authority to the pilot whenever in Bypass mode.
- e. **Bypass indication** – This determines what, if any, indications are provided to the pilot when the SAS is in Bypass Mode. When 'Chime' is selected, a chime sound will be played every couple of seconds when the cyclic input exceeds the aforementioned threshold percentage and is in fly-through mode. A single, lower frequency chime will be played when the SAS regains authority after the cyclic is returned to within the threshold and fly-through is cancelled. When 'Flashing' is selected the SAS light will flash. 'Chime + Flash' may be selected to enable both visual and audible cues.
- f. **Saturation indication** – This determines what, if any, indications are provided to the pilot when the SAS has reached the end of its authority in either longitudinal or lateral axis and is unable to exert more power to attempt to achieve the desired attitude. The user can select a combination of chimes and flashing red SAS LED.
- g. **Max cyclic pitch authority %** - This is the maximum cyclic longitudinal (pitch) deflection the SAS can exert before it becomes saturated.
- h. **Max cyclic bank authority %** - This is the maximum cyclic lateral (bank) deflection the SAS can exert before it becomes saturated.

Setting either of these too high when disabling SAS bypass mode (by setting deviation detection method to None, or setting a very high bypass cyclic threshold %), can result in the pilot not having enough cyclic authority to overcome the strength of the SAS, effectively running out of control movement.

- i. **Play Nav capture chime** – Plays a single chime when the autopilot captures the HSI's Course Deviation Indicator or Glide Slope Indicator needles.
- j. **External SAS actuators** – This is an experimental feature, whereby the SAS does not apply cyclic deviations in software, but instead outputs deviations via SimConnect that a physical force-feedback system can apply instead.
- k. **Disable SAS limiter** – When SAS speed hold mode is engaged, if the airspeed deviates more than 20 knots from that when speed hold mode was engaged, the speed mode will automatically disengage. When altitude hold mode is engaged, the mode will disengage if the altitude deviates more than 500 feet from that when the mode was engaged. (In either case, the pilot is informed via a series of audio tones.) When checked, these cut-off limiters are disabled.
- l. **Trim hat bump extended functions** – This option permits the trim hat to adjust SAS upper mode (heading, navigation OBS, speed and altitude) set-points when modes are operating.
- m. **CTS is relative** – Select this if you are using a Flight1 GPS device to match the setting in the GPS settings to ensure the autopilot steering cues are interpreted correctly. (Or leave unchecked if you are using the native platform GPS.)

- n. **CRS is linked** – Select this if you are using a Flight1 GPS device to match the setting in the GPS settings.
5. **Instrumentation** – Applies global settings to the following cockpit instrumentation settings:
- a. **Simple AI gyro** – When checked, the aircraft's Attitude Indicator gyro will spool up in 5 seconds when power is applied, as opposed to the normal 2 minutes) and runs down in 15 seconds when power is removed, as opposed to the usual 15 minutes. When checked the AI is also not subject to acceleration forces and is always caged to the true horizon.
 - b. **AI self-erects** - When checked, manual caging of the Attitude Indicator, (normally performed by pulling the AI's adjuster knob once the AI is fully spooled up), is not required as the AI will self-erect when spooled up.
 - c. **AI rigidity %** - Set a coefficient percentage that describes how stable the AI gyro remains when exposed to longitudinal and lateral acceleration forces. Lower values will cause the gyro to react and recover more slowly when the aircraft is manoeuvring.
 - d. **VFR squawk code** – Pressing the VFR button on the Transponder control causes it to set the squawk code to the default VFR value. The simulation defaults to a value of 1200 as per use in the USA. Some countries operate a different default code. (For example in the UK, a value 7000 is used.) Setting this value changes the default code applied when pressed.

Note: The built-in air traffic control (ATC) system within both FSX and Prepar3D uses a VFR squawk code of 1200 worldwide. Changing this default value is likely only of use in a training environment or within an external ATC system employing correct regional procedures, (such as VATsim.)

- e. **ADF default type** – This drop-down selection determines which type of ADF instrument is fitted by default on the aircraft's main panel. The options are:
 - i. **FCC (Fixed Compass Card)** – The compass rose is fixed and does not rotate. Hence there is no rotation knob provided. The pilot must transpose the difference between the fixed indicator heading and the aircraft's heading manually.
 - ii. **RCC (Rotatable Compass Card)** – A rotation knob permits the pilot to manually rotate the compass rose and align it with the aircraft's magnetic heading.
 - iii. **Single-needle Radio Magnetic Indicator (S-RMI)** - The compass rose rotation is driven by a gyro, but an adjuster knob is provided to permit the pilot to adjust the rose to set a required bearing and compensate for any gyro drift.
 - f. **Default Annunciator Layout** – Selects the default layout of the annunciator panel according to the various layouts specified in the `DodoSim 206.cfg` file.
6. **Aircraft Specific ({tail number})** – Applies settings to the current tail number aircraft only:
- a. **Fuel Control Unit style** – Determines the type of fuel control unit fitted to the aircraft and can be set to either of these systems:
 - i. **CECO-style** – Fuel is fully metered by the pilot during the engine start-up procedure, requiring them to gradually open the throttle during light-off and modulate the fuel input using the throttle until the turbine is self-sustaining. A turbine over-temp is much more likely due to throttle misuse when using this FCU system.

- ii. **Bendix-style** – Only enough fuel is permitted to be input until the engine is self-sustaining. A turbine over-temp due to throttle misuse is more unlikely using this system. A pilot ordinarily snaps the throttle open passed and then back to the idle stop latch rather than gradually opening it during light-off.

Note: That the previous DodoSim 206 FSX version had FCU type hard-coded into the aircraft configuration files. By adding it as an option here, flexibility was retained whilst removing the need to have twice the number of variants in the aircraft selection dialog. The FCU type is stored for the given tail current number when changed.

- b. **ADF type** – Sets the ADF type (as explained prior) for this specific aircraft tail number.
- c. **ADF Flight timer ground sense** – The ADF control incorporates a flight timer. This normally starts running from zero upwards after power is applied. It can however be paused by either a pilot input, or automatically. When checked, the flight timer will automatically be paused when the aircraft is on the ground, (as if utilising a strain gauge fitted to the landing gear.)
- d. **Has SAS fitted** – Determines whether this aircraft has Stability Augmentation System fitted or not.
- e. **Annunciator Layout** – Selects the layout of the annunciator panel according to the various layouts specified in the `DodoSim 206.cfg` file.
- f. **Reset persistent data for {tail number}** – Clicking this icon causes the persistent data, (such as cumulative wear, hours flown, number of engine starts, etc. for this aircraft to be deleted.)

Note: Persistent data, aircraft specific and general settings files are stored in the Saved Flights folder; `My Documents\{platform} Files\` (“My Documents” is named simply “Documents” in Windows 10.)

`DodoSim 206.cfg` contains the global settings data,

`DodoSim 206_as_specific.cfg` contains the settings for individual aircraft,

`DodoSim 206_{tail number}.dat` contains the persistent usage data for the given specific aircraft.

Deleting any file will cause it to be re-created when the aircraft is next loaded in the simulation platform and default values will be set. Manual editing of any configuration file should be done with great care and is discouraged.

4.2 CONTROLS SETTINGS

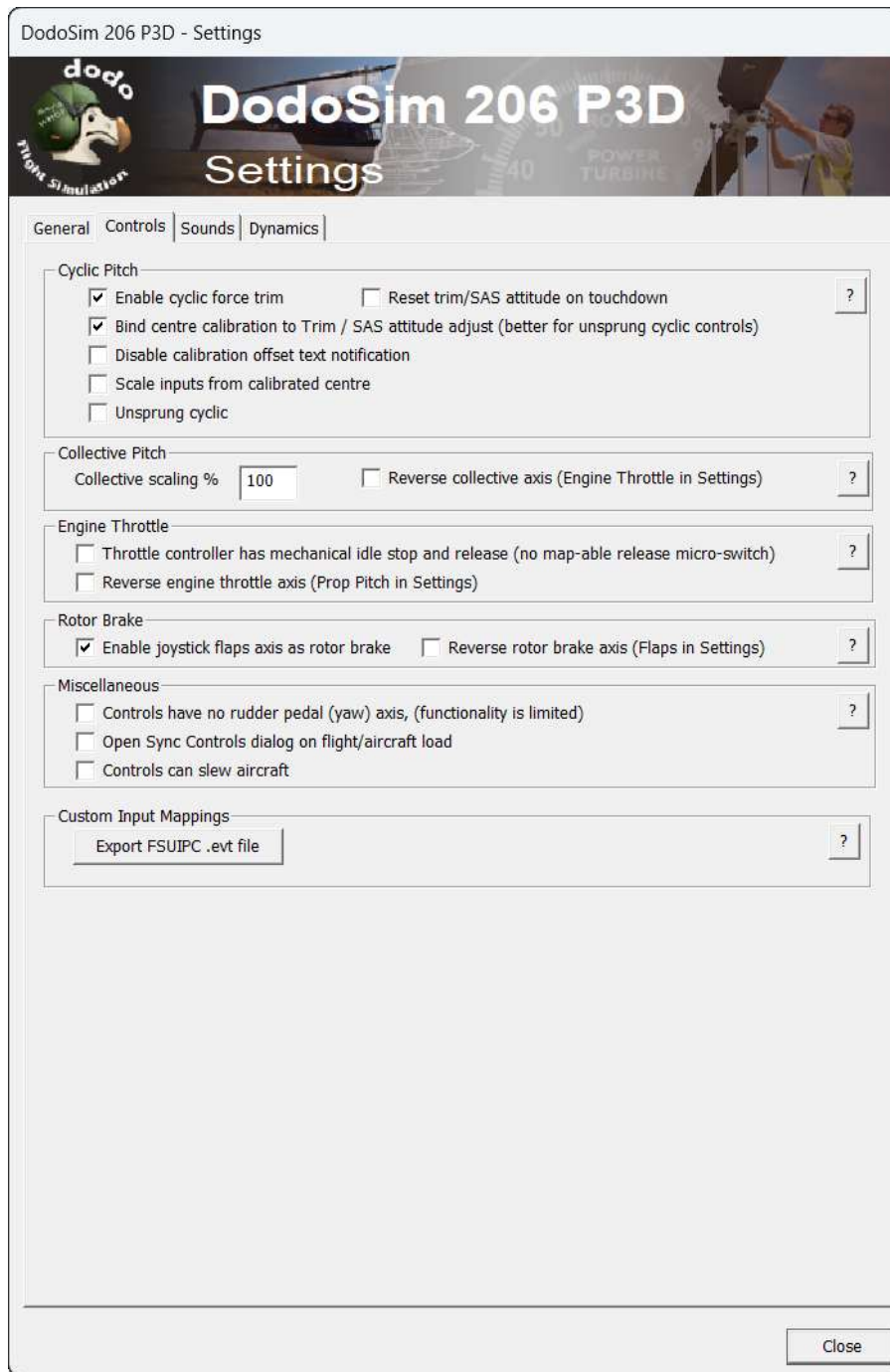


Figure 10 – DodoSim 206 V2 Controls settings

The Controls Settings dialog tab options are split into six groups:

1. **Cyclic Pitch:**

- a. **Enable cyclic force trim** – when checked the pilot is able to offset stick deflection forces using a cyclic force-trim function, where either a force cancel feature can be

invoked permitting the pilot to release tension on the stick whilst the aircraft maintains the same attitude, or trim can be “nudged” in any cardinal direction by use of four trim inputs. Note that when SAS is energised, manual trim functions are non-functional.

- b. **Reset trim/SAS attitude on touchdown** – Since it is more difficult to detect that an aircraft is out of trim in a simulator than for real, leading to sudden unwanted attitude upon take-off, checking this option ensures that any trim deflection or SAS attitude setting is automatically reset to neutral when the aircraft touches down.
- c. **Bind centre calibration to Trim / SAS attitude adjust** – This setting is only applicable for cyclic sticks that are not spring loaded or otherwise self-centring, where there is no need for a pilot to relieve pressure on the stick for the sake of comfort during extended transitions. It automatically re-calibrates the centre position of the stick (and therefore the SAS deflection threshold and trim centre) to the current position whenever the trim force release or SAS attitude set are triggered. (For non-self-centring cyclic sticks, the use of trim is itself negated but SAS is still useful.)
- d. **Disable calibration offset text notification** – When a cyclic centre calibration signal is received, the pilot is informed of the X and Y axis deflections as percentages of the measured electrical potentiometer ranges. The feature is used to reduce the difference between the resting neutral cyclic hand position and then centre of the actual electrical potentiometers measuring deflection.



DodoSim 206 P3D - Cyclic stick centre position calibrated X = +13% Y = +8%

Figure 11 – Cyclic centre calibration text notification

This gives the pilot an indication of how much travel is available in any direction. Calibrating too far from the ideal 0% positions can limit cyclic throw in the particular direction leading to a loss of cyclic authority during manoeuvring. For a resting, hovering position the cyclic should be calibrated as close to 0% as possible. For an installation using a non-self-centring cyclic, the pilot should find a comfortable resting position for the cyclic and then send the calibration signal to set the centre point, taking note of how far the adjustment is from true centre and adjust if desirable. The message will be appended with a cue indicating which way to move the stick before re-attempting calibration. The text will be colour-coded according to the severity of the deviation. Red indicates a gross deviation, yellow a moderate one and green a negligible one.

When checked the pilot will not receive a textual notification of the new calibrated centre position.

- e. **Scale inputs from calibrated centre** – Checking this option causes cyclic inputs to be scaled to their maximum value from the calibrated centre, ensuring that full cyclic authority can be maintained when the cyclic is calibrated far from the electrical potentiometers centre positions. It should be noted though that the further the centre position is calibrated from the electrical centre, the more unequal the effect of cyclic input will be to opposite sides of the calibrated centre.

2. Collective Pitch:

- a. **Collective scaling %** - This percentage permits the collective pitch axis range maximum value to be scaled for custom controls and comfort, such that the maximum height to which the collective lever need be raised to achieve maximum pitch deflection can be set. A higher value than the default 100% will ensure maximum deflection is reached at a lower lever position and the therefore collective pitch response will more sensitive at lower physical settings. Note that raising the scaling value reduces the

resolution of inputs and so some finer granularity of control is lost in favour of smaller physical deflections.

- b. **Reverse collective axis** - Reverses the axis input assigned to collective pitch for the DodoSim 206 V2 only. (All other aircraft within the simulation remain unaffected.)

3. Engine Throttle:

- a. **Collective throttle has mechanical idle stop and release** – Permits calibration and use of a mechanical idle stop (if the physical throttle is fitted with one) to limit throttle travel to prevent accidental engine shut-down.

Selecting this option will open a dialog asking whether the idle stop position should be calibrated:

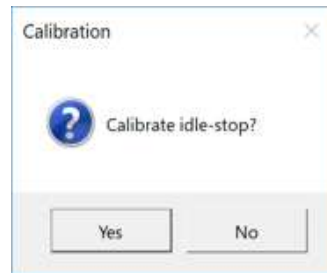


Figure 12 – Idle stop calibration query

If “Yes” is selected the dialog will be replaced with another announcing that the simulation will run for 5 seconds during which time the throttle should be opened past and then back to rest upon the idle stop position:

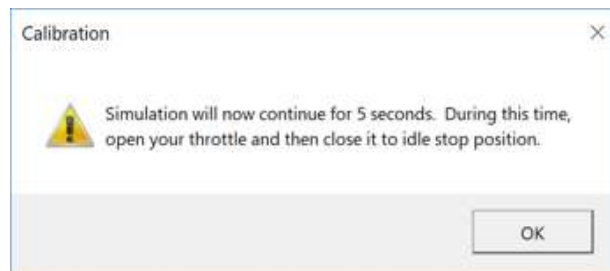


Figure 13 – Idle stop calibration instruction

“Ok” should be selected, returning to the sim, where input is suspended. Set the throttle to rest upon the mechanical idle stop position and wait for the 5 seconds to expire. At this time the Settings dialog will re-open and a dialog will be shown indicating the axis potentiometer value read at which the idle stop is located. Do not be concerned about the actual value reported as it is very dependent on the nature of the potentiometer used in the construction of the throttle, unless it reads as 0%, in which case it is likely that there is a dead zone below which the potentiometer cannot be read and this extends into the idle stop range. It is possible that the throttle is faulty or in need of calibration within Windows.

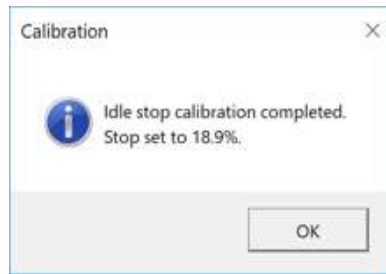


Figure 14 – Idle stop calibration result

The throttle axis is subsequently calibrated either side of the measured idle stop position to give an opening value of 6% throttle at the idle stop position.

Un-checking this option disables any previous calibration. If the box is checked and “No” is selected in response to the request to calibrate, then the previous calibration result will be used. (Or none if no prior calibration has been performed.)

Caution: Selecting this option when using a throttle without a no physical idle stop latch will result in the throttle being able to be closed all the way, activating the idle stop in the software and shutting down the engine.

- b. **Reverse collective axis (Engine Throttle in Settings)** - Reverses the axis input assigned to engine throttle for the DodoSim 206 V2 only. (All other aircraft within the simulation remain unaffected.)

4. Rotor Brake:

- a. **Enable joystick flaps axis as rotor brake** – Setting this option binds the flaps axis in the simulator to a proportional rotor brake. The brake can then be applied to slow the rotor after the engine shut down. In order to determine if this axis requires reversing, look upwards in the virtual cockpit and manipulate the axis. When the lever is in the down position, the brake is applied. A squealing noise will be heard if the brake is applied when the engine is running (or if applied too firmly when the rotor is running quickly.)
- b. **Reverse rotor brake axis** - Reverses the flaps axis input assigned to the rotor brake for the DodoSim 206 V2 only. (All other aircraft within the simulation utilising the flaps axis remain unaffected.)

5. Miscellaneous:

- a. **Controls have no rudder pedal** – Setting this option disables torque-induced yaw and yaw axis cross-couplings for use in set-ups that have no rudder pedals, significantly limiting realism.
- b. **Sync controls on load** – For hardware toggle switches in hardware cockpits, this displays a dialog showing all cockpit switch positions applicable to the starting condition of the aircraft when a flight is loaded to permit re-synchronisation of hardware switches before the flight commences:



Figure 15 – DodoSim 206 V2 Controls Sync dialog window

Note for hardware cockpit builders: If your hardware cockpit is missing some of these functions and you wish to set them permanently to a specific state and remove them from this dialog, you can do so by un-commenting the respective “EXCLUDE” line (remove the \\ preface) in the DodoSim 206.cfg file, located in the saved flights directory. Then load a flight, open all the 2D panels, set the switches as required and save the flight. You can then use this flight as the basis for all flights or training scenarios by simply loading it, moving the aircraft to the desired location, setting the environment as required and re-saving it.

- c. **Controls can slew aircraft** – Un-checking this option prevents aircraft controls from moving the aircraft in slew mode.

6. Custom Input Mappings:

This group contains a single item:

- a. **Export FSUIPC .evt file** – This function writes a text file to the {application root}/Modules/DodoSim 206 V2.evt directory containing a list of all custom events supported by the DodoSim 206 V2 which a registered version of FSUIPC is able to bind hardware inputs to. Binding events in this manner is described later in this document.

4.3 SOUNDS SETTINGS

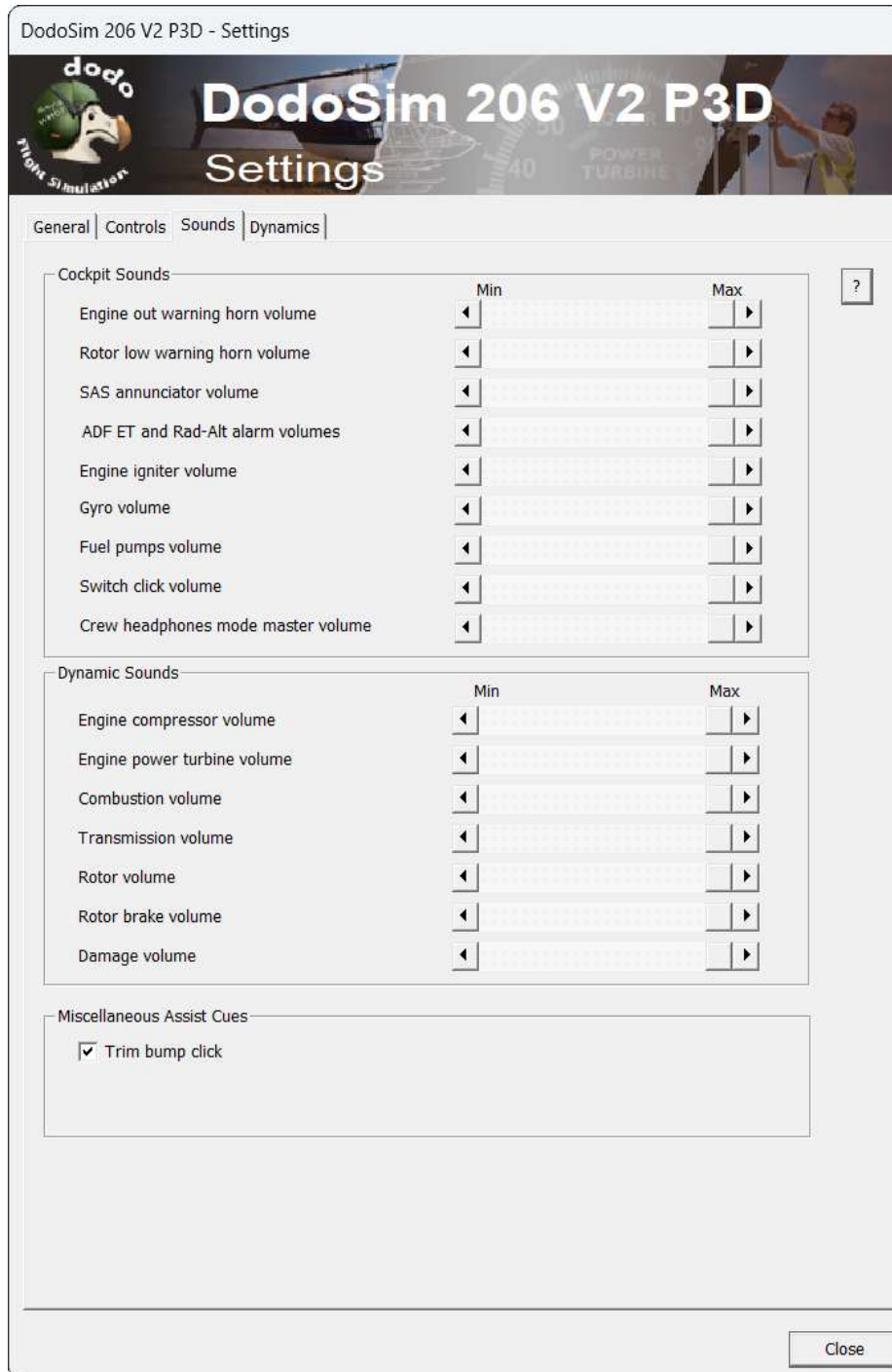


Figure 16 – DodoSim 206 V2 Sounds settings

The Sound Settings dialog tab contains slider controls used to set the relative volumes of the sounds produced by the DodoSim206 that are played by the aircraft software rather than the native simulation.

The sliders are arranged in three groups:

1. **Cockpit sounds** – These are only audible when inside the aircraft and are muted when using an external view. They are static sounds, meaning they do not vary in frequency or volume depending on any simulation parameter.
2. **Dynamic Sounds** – These sounds vary in pitch and volume depending on simulation parameters, i.e. engine or rotor speed. They are audible from both with and outside the aircraft.
3. **Miscellaneous Assist Cues** – These sounds are provided as additional cues to the pilot to indicate when an input or event occurs.

The sliders are self-explanatory with the exception of “Crew headphones mode master volume”. When headphones mode is enabled, in many case sounds are decreased in volume and in some cases alternative filtered sounds are played in their stead. Manipulating this slider applies a blanket volume adjustment to all the sounds.

The “Trim bump click” option emits a click sound when either a cyclic trim adjustment, SAS attitude nudge, or SAS upper mode speed or altitude set-point adjustment is made.

Enabling “Trim bump click” is useful to gauge and moderate the size of the change required as these functions can be set to rapidly repeat when the event-mapped button is held depressed.

The settings within this dialog relate to the DodoSim 206 V2 aircraft only and have no impact or applicability to other aircraft supplied with Prepar3D/Microsoft Flight Simulator X or by any other third-party.

4.4 DYNAMICS SETTINGS

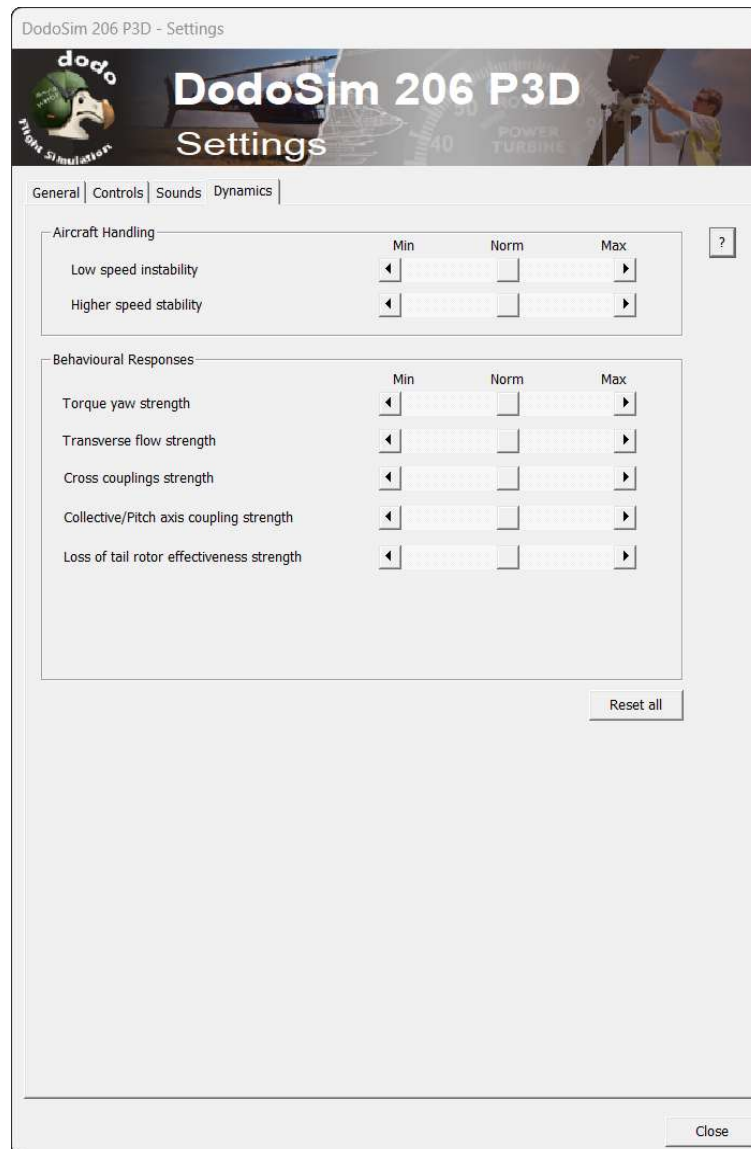


Figure 17 – DodoSim 206 V2 Dynamics settings

The Dynamics Settings dialog tab contains global settings that moderate the aircraft's behaviour. The purpose of these is to allow a pilot to fine tune their experience. Experienced real-world helicopter pilots might be put off by having to consciously account for behaviours in the simulation that they would naturally “tune out” and account for subconsciously in a real aircraft due to non-visual motion cues, such as inner-ear and other bodily sensory mechanisms. Simulation pilots wanting a more challenging experience might choose to increase aircraft instability and cross-couplings.

Configuration options are divided into two sections:

1. **Aircraft Handling** – These govern the aircraft's stability under different flight regimes:
 - a. **Low speed instability** – This dictates how unstable the aircraft is at low speeds. Increasing the value will require the pilot to make greater continuous cyclic inputs to maintain a level attitude. The low speed instability coefficient decreases as airspeed increases.

- b. **High speed stability** – This control determines the amount by which the aircraft gains attitude stability as the airspeed increases. Increasing the value will increase the aircraft's tendency to trend towards a neutral attitude if pressure on the cyclic is released, requiring the pilot to maintain cyclic deflection to maintain a desired non-neutral attitude.
2. **Behavioural Responses** – These govern to what extent power and control demands effect pitch, roll and yaw moments of the aircraft.
 - a. **Torque yaw strength** – This scales the effect of tail rotor application has on the torque-induced yaw effect. Increasing this value will increase the amount of anti-torque pedal input is required to counter the torque-induced yawing moment.
 - b. **Transverse flow strength** – Also known as inflow roll, this value determines how much tendency the airflow into the rotor causes the aircraft to roll (in forward flight) due to the dissymmetry of lift between the advancing and retreating blades. Increasing this value will increase this effect, requiring greater lateral cyclic input to counter it as airspeed increases.
 - c. **Cross-coupling strength** – This setting dictates to what extent moments on one aircraft axis affect those of another, for example, yaw (pedal) input is cross-coupled with roll because the main and tail rotors do not lie on the same axis. (One reason why some helicopters elevate their tail rotor to bring its hub closer vertically to the main rotor hub is to counter this tendency by design.) The higher the value, the more the pilot will have to compensate for movements on one axis with control inputs on another.
 - d. **Collective/Pitch axis coupling** – This setting determines to what extent increases in collective input effect the pitch axis. At higher values, increasing the pitch would require the pilot to apply slight forward cyclic to prevent the nose from rising.
 - e. **Loss of tail rotor effectiveness strength** – The setting determines the extent to which tail rotor effectiveness is lost when either it is ingesting main rotor wash, or contending with a side-wind. Higher values will require greater care and control in susceptible conditions and may result in a complete loss of effectiveness in extreme circumstances.

4.5 LOCATION OF SETTINGS FILES

The DodoSim 206 V2 stores its settings files in the

'C:\Users\{username}\Documents\{platform} Files' directory.

- DodoSim 206.cfg – Contains generic settings..
- DodoSim 206_ac_specific.cfg - Aircraft specific settings, i.e. equipment load-out.
- DodoSim 206 V2_{tail number}.dat – Contains persistent usage and damage information (for when in "Cumulative Wear" mode.

Deletion of any one of these files will return all the relevant parameters contained within to their default values and force the file to be re-written with default values the next time the aircraft is loaded in the simulator.

4.6 ADDITIONAL SETTINGS NOT ACCESSIBLE VIA THE DIALOGS

This section describes additional settings that can only be adjusted outside of the simulation platform application by manual editing of the 'DodoSim 206.cfg' text file.

Caution should be exercised when editing this file. Mistakes may cause undefined behaviour and potentially cause the software to crash when the aircraft loads. Should this occur, simply delete the file and it will be re-created with default values when the aircraft is next loaded in the simulator.

Manual changes to the file can be enacted easily in the simulator without resorting to restarting it, or reloading the aircraft or scenario, by using FSUPIC to map the DS206.ZDEBUG RELOAD SETTINGS event to a key or joystick button. (See section 5.1.2 Mapping using the FSUIPC .Evt File.)

4.6.1 Pop-Up Window Screen Positioning

The Settings, Failures and Controls Sync dialog windows will open centred in the main simulation window by default. It may be that the lower portion of this window is obscured from a pilot's seated position when running in a physical cockpit environment and using a large floor to ceiling display.

If one of these windows is moved at any time using the mouse then the position will be saved for all windows in the future.

The screen position of the top-left of the pop-up windows can be adjusted manually within the DodoSim 206 V2.cfg file, located within the saved flights directory, (\{user}\Documents\{platform} Files\). This file can be edited in any Windows text editor, such as Notepad.

Within the file are two entries, as below:

```
POPUP_WINDOW_X_POS=0
POPUP_WINDOW_Y_POS=0
```

The value 0 will cause the X or Y position to be centred within the main Prepar3D application window. Any other value is relative to the top/left of the window. For example, setting POPUP_WINDOW_Y_POS=30 will cause the window to open near the top of the display.

If changed while the DodoSim 206 V2 is running within the simulation, the aircraft or flight should be reloaded to ensure the changes take effect.

4.6.2 Creating New Annunciator Layouts

Within the 'DodoSim 206.cfg' file, under the 'Instruments' section, there exists (by default) two annunciator configurations, labelled 'Standard' and 'Alternative'.

Additional layouts can be added by the addition of a new configuration line, i.e.:

```
ANNUNCIATOR_CONFIG_DEFINITION=ConfigLabel
1,EO,TP,SP,EC,SP,FF,SP,SP,SP,IP,RL,TT,SP,TC,TR,FP,SP,SP,GF,BD
```

Where *ConfigLabel* is the name that appears in the Settings dialog for the user to select. (Configuration labels should not include spaces.)

Following the label, there are twenty abbreviations detailing which annunciator lights appear at each of the twenty positions on the annunciator, starting from the top-right to top-left, followed by bottom-right to bottom left.

i.e. The Standard layout is defined as:

```
ANNUNCIATOR_CONFIG_DEFINITION=Standard,EO,TP,BH,EC,BD,FF,GF,SP,SP,IP,RL,TT,B
T,TC,TR,FP,SP,SP,FL,SP
```

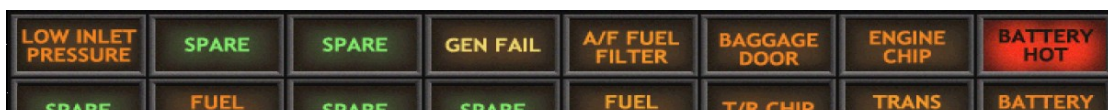


Figure 18 – ‘Standard’ annunciator layout

Available indicator abbreviations are:

Abbreviation	Meaning	Colour
EO	Engine Out	Red
TP	Transmission Oil Pressure Low	Orange
EC	Engine Chip	Orange
FF	Fuel Filter Contaminated	Orange
FP	Fuel Pump Fail	Orange
FL	Fuel Quantity Low	Orange
RL	Rotor RPM Low	Orange
TT	Transmission Oil Temperature High	Orange
TC	Transmission Chip	Orange
TR	Tail Rotor Chip	Orange
GF	Generator Fail	Yellow
BD	Baggage Door	Orange
BH	Battery Hot	Red
BT	Battery Temperature Warm	Orange
IP	Inlet Pressure Low	Orange
SP	Spare	Green

Table 2- Available annunciator light configuration abbreviations

Indications may be duplicated, (as is Spare in the Standard layout example above). It is not possible to add new indications, or change the colours of the supported ones.

4.6.3 Control Synchronisation Exclusions

If the setting “Open Sync Controls dialog on flight/aircraft load” is selected in the Settings->Controls tab, then a window is opened every time a new flight is loaded or the aircraft is re-loaded. While this window is open, all switch input is prohibited from reaching the simulator. This permits users of hardware cockpit switches to reconfigure them to match the states listed in the dialog without these actions having an impact on the simulated aircraft before closing the windows and continuing with the flight.

The list provided is an exhaustive list of all cockpit switches. Users operating a partial hardware cockpit may reduce this list for brevity. To do so, remove the \\ comment indicators from the front of the control within the ‘Sync Controls dialog list exclusions’ section, i.e. Removing the \\ from ‘\\EXCLUDE_SYNC_CONTROL=GEN FIELD’ omits the Generator Field circuit breaker from the Sync Controls dialog.

4.6.4 Additional SAS Configuration Customisation

4.6.5 Individual Settings

There are additional configurable options for the SAS in the settings file that are not included in the Settings dialog:

DISALLOW_AP_GPS_DRIVES_NAV_SET – The DodoSim 206 V2 has a mechanical switch on the main panel for switching the HSI input between the Nav 1 radio and GPS. Some third-party GPS systems (like the Flight1 or Reality XP systems) can provide this via their touch-screen functions (CDI icon switches GPS/VLOC modes). If the user is operating a full hardware cockpit with a physical switch, then the GPS can be prohibited from performing this function, relying instead on the pilot to set the switch accordingly.

SAS_POST_DURATION – Specified the number of seconds that the SAS performs its Power-On Self-Test sequence for when powered up.

`SAS_ENABLE_HDG_NAV_PEDAL_AXIS_ASSIST` – Intended to permit pilots with centre-sprung pedals to remove their feet from them when the Autopilot is operating in heading or navigation hold modes, this setting applies PID controller assistance to the pedal axis to ensure the flight remains coordinated.

`SAS_RESET_ATTITUDE_PIDS_IN_FLYTHROUGH` – This is an experimental feature that should not be required. It resets the amassed SAS attitude PID values when the SAS is operating in Bypass Mode.

`SAS_DISABLE_SAS_BYPASS` – This is a quick way of completely disabling entry into SAS bypass mode by cyclic stick deflection. Regardless of other settings, (such as the deviation detection mode), setting this to YES will prevent the SAS from entering bypass mode upon cyclic deflection, resulting in the pilot being forced to “fly through” the SAS inputs, (unless the Force Trim Release (cyclic trim) / SAS Attitude Set button is depressed).

`SAS_PITCH_MAX_DEGREES` – The maximum (positive and negative) pitch the SAS is permitted to drive the aircraft to.

`SAS_BANK_MAX_DEGREES` – The maximum (positive and negative) bank the SAS is permitted to drive the aircraft to.

`SAS_LOCK_INTEGRALS_ON_SATURATION` – This determines whether the PID integral values should be frozen when the SAS reaches maximum authority and saturates.

`SAS_LOW_SPEED_THRESHOLD` – This is the minimum airspeed that the aircraft can enable and perform autopilot functions of heading, navigation, speed, altitude and vertical hold. Forward flight is required for these modes to operate effectively.

`SAS_TRIM_HAT_DUTY_CYCLE_SEC` – This determines the frequency at which repeated trim hat events are permitted to be actioned, (with the exception of extended functions for adjusting the heading bug and HSI OBS needle.) (Mechanical Cyclic Trim adjustments are not subject to the duty cycle.)

`SAS_TRIM_HAT_NUDGE_DEGREES` – The amount by which the SAS attitude hold set-point is adjusted by at each duty cycle event.

`SAS_CYCLIC_SLOP_COMPENSATION` – When set to YES, this setting attempts to account for small movements in the cyclic when exiting bypass mode. This is useful if the cyclic stick is large and has a small amount of slop in its mechanism.

`SAS_PITCH_OVER_ROLL_COMPENSATION_FACTOR` – This coefficient determines how much pitch adjustment the SAS makes when it is performing a banking action, in order that the nose does not drop.

`SAS_ATT_HOLD_NOSE_DOWN_LIMIT_DEGREES` – This is the nose down pitch limit of the SAS attitude hold mode.

`SAS_ATT_HOLD_NOSE_UP_LIMIT_DEGREES` – This is the nose up pitch limit of the SAS attitude hold mode.

`SAS_ATT_HOLD_BANK_LIMIT_DEGREES` - This is the bank limit of the SAS attitude hold mode.

`SAS_VOR_STEERING_SENSITIVITY` – This coefficient determines the sensitivity of the autopilot steering to the HSI Course Deviation Indicator when tuned to a VOR transmitter.

`SAS_VOR_STEERING_MAX_MAGNITUDE_DEVIATION_DEGREES` – This coefficient determines the sensitivity of the autopilot steering to the HSI Course Deviation Indicator when tuned to a VOR transmitter.

`SAS_VOR_STEERING_GRAB_DEVIATION` – This value determines the percentage of HSI OBS needle deviation at which the autopilot navigation hold mode will latch and become driven by the deviation when tuned to a VOR transmitter.

`SAS_LOCALISER_STEERING_SENSITIVITY` - This coefficient determines the sensitivity of the autopilot steering to the HSI Course Deviation Indicator when tuned to an ILS transmitter.

`SAS_LOCALISER_STEERING_GRAB_DEVIATION` - This value determines the percentage of HSI OBS needle deviation at which the autopilot navigation hold mode will latch and become driven by the deviation when tuned to an ILS transmitter.

`SAS_SPEED_DEV_CUT_OUT` - The deviation (in knots) between the current airspeed and the speed hold mode set-point at which the autopilot drops out of speed hold mode.

4.6.6 Configuring the PID Controllers

Proportional, Integral and Derivative controllers are a staple of mechanical automation. They compare the difference between two values, a current value and a desired set-point, and perform a mathematical function to apply a proportional power adjustment to reduce difference between them. This control loop is run frequently to account for changes in the values.

If that was all they did then they would be considered an “open-loop” system, where there is no feedback. PIDs introduce two additional factors to the Power component; an Integral component, which is an accumulation of all errors since the loop started running, and a Derivative component, which is calculated as the difference between the deviation this time and the last time the loop was run.

The mathematics of a basic PID loop is very simple, but very effective and can be written as:

```
P = (target - current)
I = (last I + P)
D = (P - last P)
Output = Sum of ( (P * pC * dT) + (I * iC) * dT) + (D * dC * dT) )
```

Where:

`pC`, `iC` and `dC` are coefficients, tuned to produce the best response minimising undershooting, and overshooting, causing “porpoising”.

`dT` is the difference in time since the last time the loop was run.

The PID controller functions in the DodoSim 206 V2 implement additional features depending on their purpose. Some require limits on their integral values, to limit the cumulative error build up when correction to the desired value may take a significant amount of time, (such as turning to a desired distant heading). Some have integral windows around the target value within which the integral is allowed to operate, otherwise it is discounted from the equation.

Autopilots operate inner and outer control loops. The SAS attitude mode operates only two loops, one for pitch and the other for bank. The higher modes operate additional loops around those. For example, the heading hold mode has an outer PID control loop driven by the heading, and its power output (a driven from the deviation between the current heading and the desired heading) is used to set a desired bank angle into the bank attitude PID controller. Similarly, the speed and altitude modes have their own outer loops driven by speed and altitude deviations respectively, and both utilise the pitch PID controller loop.

Some systems have a bias, where a constant tweak is required to be applied to the power output.

The DodoSim 206 V2's PID loops are described in the settings file using four parameters, for example:

```
SAS_ALTITUDE_PID=0.0170,0.0040,0.0600,0.0150
```

The four parameters are the Proportional, Integral, Derivative and Bias coefficients.

The DodoSim 206 V2 PIDs have been tuned through extensive testing to provide the best performance, aiming to achieve the pilot desired set-points as quickly as possible, within comfort constraints, whilst limiting tendencies for undershoot, overshoot and porpoising. Whilst it may be possible to achieve even better performance with minor tweaking, it is not recommended. Undesirable flight behaviour may result. Small changes can result in large differences in behaviour.

Should mistakes be made, the default values can be recovered by deleting parameter lines from the settings file, and opening and closing the Settings Dialog in the simulator, at which point they will be rewritten to the file.

4.6.7 Debugging Options

SETTINGS_DEBUG_OUTPUT_CONTROLS – Setting this will add a debug text block to the on-screen notification detailing control inputs and responses:

```
DodoSim 206 P3D - Debug Data:
[DEBUG_OUTPUT_CONTROLS]
Raw: CyX = 0.000%, CyY = 0.000%, Ped = 0.000%, Col = 45.389%, Thr = 0.000%, RBrake = 0.000%
Factored: CyX = 0.000%, CyY = 0.000%, Ped = 0.000%, Col = 45.389%, Thr = 0.000% (Display pos = 0
```

Figure 19 – Controls debug output

The Raw values indicate the axis values detected by the simulator prior to any filtering, scaling, reversals or detent locks (for the throttle axis). Cyclic and Pedal inputs range from -100% to +100%, throttle, Collective, Throttle and Rotor brake (flaps axis) axes range from 0 to +100%.

DEBUG_OUTPUT_SAS – Setting this will add a debug text block to the on-screen notification detailing the SAS status.

```
DodoSim 206 P3D - Debug Data:
[DEBUG_OUTPUT_SAS]
SAS: Actuator Pwr = On, Control Pwr = On
Modes: ATT = On, HDG = Off, NAV = Off, SPD = Off, ALT = Off, VRT = Off
Pitch Target = 0.6 (current = 0.6), Bank Target = 0.0 (current = -0.2), Pitch Actuator = 0.0%, Bank
Cyclic bypass sensing method: Deflection, limit = 15%, current X = 0.0%, current Y = 0.0%, Trim s
```

Figure 20 – SAS debug output

The SAS debug output will state the POST status, current modes, attitude, speed, altitude and nav heading settings, and indicate when the actuators are not functioning when the SAS is in bypass mode:

```
DodoSim 206 P3D - Debug Data:
[DEBUG_OUTPUT_SAS]
SAS: Actuator Pwr = On, Control Pwr = On
Modes: ATT = On, HDG = Off, NAV = Off, SPD = Off, ALT = Off, VRT = Off
Pitch Target = 0.6 (current = 0.6), Bank Target = 0.0 (current = -0.2), Pitch Actuator = OFF, Bank
Cyclic bypass sensing method: Deflection, limit = 15%, current X = 0.0%, current Y = 0.0%, Trim s
***BASS** : Trim Release Switch
```

Figure 21 – SAS status debug output

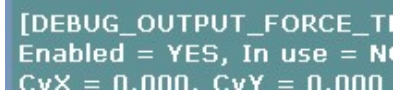
DEBUG_OUTPUT_SAS_PIDS – Setting this will add a debug text block to the on-screen notification detailing the status of the SAS PID (Proportional, Integral and Derivative) controller status and coefficients:

```
DodoSim 206 P3D - Debug Data:
[DEBUG_OUTPUT_SAS_PIDS]
PITCHPID: Got coeffs = 1, init = 0, paused = 0, p = 0.0135 (last = 0.0135), i = 0.2077 (limit = none), d = -0.0035, output = 0.18
BANKPID: Got coeffs = 1, init = 0, paused = 0, p = -0.0007 (last = -0.0007), i = 0.0111 (limit = none), d = -0.0008, output = 0.0
ALTITUDEPID: Got coeffs = 1, init = 0, paused = 0, p = 1.9516 (last = 1.9516), i = 15.0000 (limit = 15.0000), d = -0.3174, output
ALTITUDEPITCHPID: Got coeffs = 1, init = 0, paused = 0, p = -0.0297 (last = -0.0297), i = 0.0518 (limit = none), d = 0.1579, out
```

Figure 22 – SAS PID debug output

SAS PID coefficients can be modified from within the 'DodoSim 206.cfg' file, though extreme care should be taken. Even minor adjustment can result in undesirable flight behaviours, such as pitch porpoising (over-correcting), or failure to achieve the desired heading, speed or altitude hold target. (See section 4.6.6 Configuring the PID Controllers.)

`DEBUG_OUTPUT_FORCE_TRIM` – Setting this will add a debug text block to the on-screen notification detailing the status of the Force Trim system. (This only reads "In use" when it is enabled on the Controls Settings dialog tab and the SAS is unpowered.)

A screenshot of a debug text block with a dark blue background and white text. The text is arranged in three lines: the first line is the variable name in brackets, the second line shows the status of 'Enabled' and 'In use', and the third line shows the values for 'CvX' and 'CvY'.

```
[DEBUG_OUTPUT_FORCE_TRIM  
Enabled = YES, In use = NO  
CvX = 0.000, CvY = 0.000
```

Figure 23 – Force Trim debug output

5 CONTROLS CONFIGURATION

5.1 PROPORTIONAL ANALOGUE CONTROLS

The DodoSim 206 V2 is designed to be interacted with via analogue joystick controls for operating the cyclic and collective pitch, anti-torque pedals and engine throttle. Additionally, a spare axis can be used to apply proportional rotor brake if desired.

Analogue joystick axis should be mapped according to the table in section 3.3.

Cockpit Switch controls

Cockpit switch controls can be operated using the mouse left-button on both the flat 2D panels and within the 3D virtual cockpit.

Alternatively, these functions can be mapped to joystick buttons or other hardware input device (HID) controller functions.

There are two methods by which these events can be mapped:

1. Using the simulation's "Settings – Buttons and Keys" dialog window.
2. Using a registered version of FSUIPC and the included DodoSim206.evt file, (which contains names of custom events.)

Due to the high number of custom switches and buttons provided by the DodoSim 206 V2 that are not natively supported by the simulation and a shortage of unused input functions to re-purpose, use of a registered version of FSUIPC to map custom events is required if the user wishes to drive all cockpit functions from hardware, (as opposed to clicking on-screen using the mouse.)

5.1.1 Mapping events using the simulation's Settings – Controls dialog

Where features exist for which the simulation does not have an appropriate equivalent event defined, some alternative existing events are re-purposed for use within the DodoSim 206 V2. These functions act as toggles, or cycle through each switch position upon repeated presses.

The events to map to each switch function are as follows:

Function	Simulation Key Binding
Engine Starter Button	Starter 1 (on/off)
Idle Release Latch	Smoke system (on/off)
Aft Fuel Pump Circuit Breaker	Concorde nose visor (increase angle)
Fore fuel Pump Circuit Breaker	Concorde nose visor (decrease angle)
Master Battery	Master battery (on/off)
Generator Switch	Generator/Alternator (on/off)
Generator Field Circuit Breaker	APU/Generator (toggle)
Generator Reset Circuit Breaker	APU (off)
Avionics Switch	Avionics (on/off)
Anti-Collision Lights Switch	Strobe Lights (on/off)
Position Lights Switch	APU Start
Instrument Lights Switch	Panel lights (on/off)
Pitot Heat Switch	Pitot heat (on/off)
Fuel Valve	Engine 1 fuel valves (open/close)
Hydraulics Switch	Concorde nose & visor (retract)
Caution Lights Bright/Dim	Concorde nose & visor (extend)
Anti-Ice Switch	Carb heat/eng anti-ice (on/off)

GPS / Nav Switch	Nav 1/GPS (toggle)
Dir / Gyro Switch	Cowl flaps (close incrementally)
Warning Mute Button	Tail wheel (lock/unlock)
Over-Temp Test Button	Ballast valve (open/close)
Annunciator Test Button	Tail hook (up / down)
HTR Rotary	Spoilers (arm), Spoilers (extend/retract)
Caution LT (Annunciator) Circuit Breaker	Master Ignition (on)
Force Trim Release (cyclic trim) / SAS Attitude Set	Aileron trim (center)
SAS Disconnect	Autopilot master (on/off)
SAS Attitude Hold toggle	Autopilot attitude hold (on/off)
SAS Heading Hold toggle	Autopilot heading hold (on/off)
SAS Nav Hold toggle	Autopilot Nav 1 hold (on/off)
SAS Speed Hold toggle	Autopilot airspeed hold (on/off)
SAS Altitude Hold toggle	Autopilot altitude hold (on/off)
SAS Vertical (Glideslope) Hold toggle	Autopilot approach hold (on/off)
SAS attitude adjust / Cyclic trim left	Aileron trim (left)
SAS attitude adjust / Cyclic trim right	Aileron trim (right)
SAS attitude adjust / Cyclic trim forward	Elevator trim (down)
SAS attitude adjust / Cyclic trim back	Elevator trim (up)

Table 3- Default re-used or re-purposed key bindings

Some cockpit functions are momentary in nature, such as the Starter button. Unlike a switch, a momentary button returns to its previous state when released. (i.e. It only remains “on” so long as the pilot holds it down.) When momentary buttons are mapped to a joystick or cockpit button, the key repeat function should be set to maximum in order that the button is held down as long as the input is depressed, i.e.:

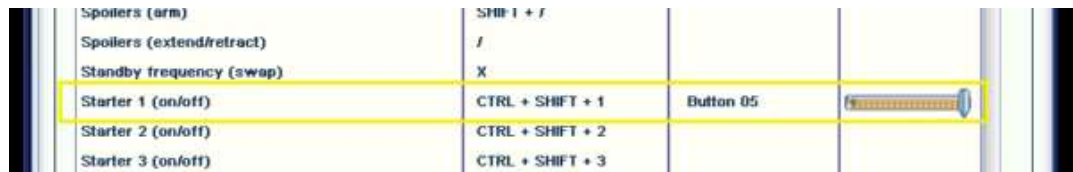


Figure 24 – Simulation's control mapping dialog window illustrating repeating function

Momentary buttons that should be set up in this way are:

Engine Starter Button
Warning Mute Button
Over-Temp Test Button
Annunciator Test Button
Force Trim Release (cyclic trim) / SAS Attitude Set
SAS Disconnect
SAS attitude adjust / Cyclic trim left
SAS attitude adjust / Cyclic trim right
SAS attitude adjust / Cyclic trim forward
SAS attitude adjust / Cyclic trim back

Table 4- Momentary functions

Important note: Note that if you map a keystroke to the Engine Starter Button function rather than a joystick button, then pressing any other key whilst you are holding down the starter key will stop the

starter key events from continuing and the engine may "hot start". (See Starting Procedure section.) You should either map a joystick button to the Engine Starter Button function, or remember not to press any other keys whilst holding the starter key down. This is due to the way in which keyboard matrices function.

5.1.2 Mapping using the FSUIPC .Evt File

Registered versions of FSUIPC include a facility to add custom events to the list of native simulation events which can be mapped to input devices. These functions are defined in the “{simulation path}\modules\DodoSim206.ev” file that is installed along with the DodoSim 206 V2 aircraft.

Important note: The DodoSim 206 V2 expects the FSUIPC.dll to reside in this modules folder. If it has been installed elsewhere, copy the .evt file to that location.

For reference, the contents of the file are listed in section 9 Appendix 1 – DodoSim 206 V2.evt File Contents.

When the simulation is started, FSUIPC may pick up the DodoSim206.evt file automatically and add the events to its list. If it does not, you can add a reference to it within the "{Simulation path}\Modules\FSUIPC.ini" file at the bottom like this:

```
[EventFiles]
0=DodoSim206
```

To map any of these functions, open FSUIPC in the simulator, select the “Buttons + Switches” tab, press the button/switch you wish to map, check the “Select for FS control” check-box and select the function from the drop down menu.

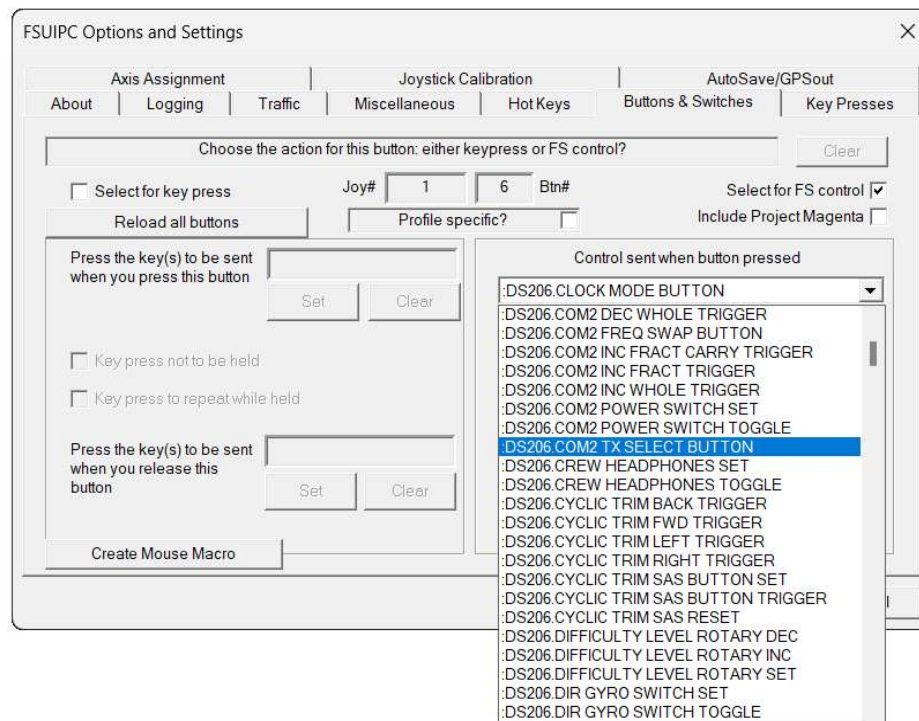


Figure 25 – DodoSim 206 V2 custom events drop-down in FSUIPC

All custom events applicable to the DodoSim 206 V2 are listed in FSUIPC preceding with “:DD206.”.

Note that in many cases custom events replicate the functionality of default functions where the platform supports the switch function natively.

Events ending with _SET require a parameter. The range of permissible values is described in the DodoSim 206 V2.evt file. For two way switches, 0 is off and 1 is on.

Events ending with _TRIGGER are momentary functions and should be set to repeat as long as the button remains held.

Events that end otherwise are single-shot functions that require no parameter and option to repeat whilst held.

For momentary button, it is advisable to use _SET functions with parameters 1 and 0 upon set and release respectively instead of a _TRIGGER alternative, as shown below:

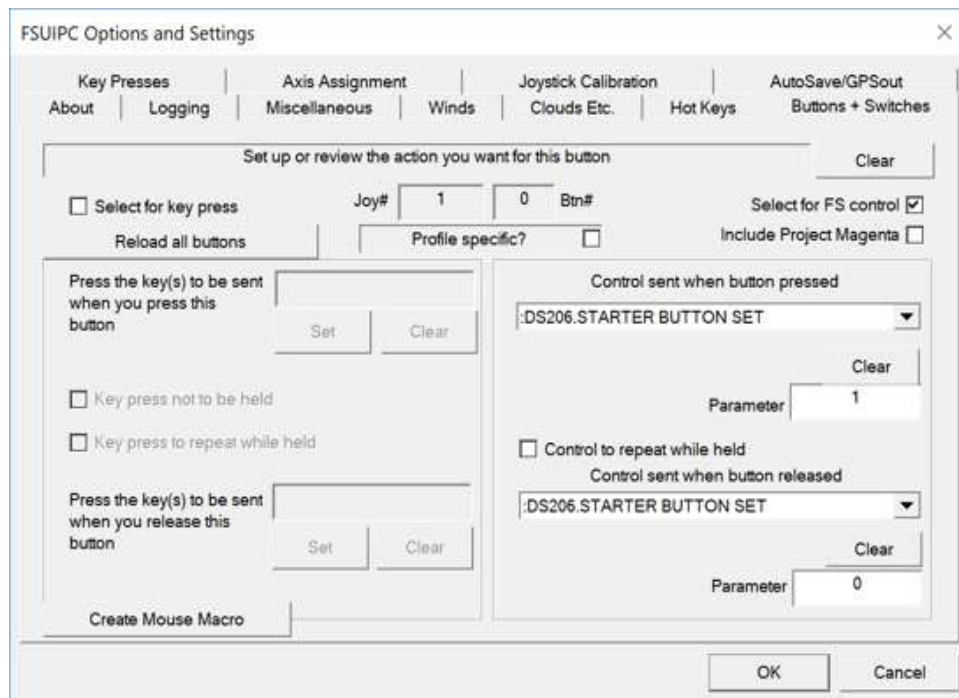


Figure 26 – FSUIPC Button + Switch dialog window showing custom events

Note that if you set both the parameters 1 and 0 for on and off, there is no need to set the control to repeat while it is held. The appropriate event will be sent when the switch/button changes state.

Note that it is advisable that events only be mapped using one of the above methods. i.e. If you map the fore fuel pump switch using the FSUIPC DodoSim206.evt event name DODOSIM206.ELECT_FUEL_PUMP1_SET, then you should not map the same switch to the “Concorde nose visor (decrease)” simulation event.

It is advisable to take advantage of the “Sync controls on load” option in the DodoSim 206 V2 Settings dialog to aid returning hardware switches to their correct state every time a flight is loaded or reset.

Important note: Since FSUIPC arranges the events listed within the DodoSim206.evt file into alphabetical order within its drop-down list, and then stores bindings according to the list index value, if this list is ever added to in a future update, you will be required to clear and re-apply all the custom

bindings, else switches will cause the wrong events to be sent.

The following sub-sections describe the custom events applicable to each switch function on each of the individual aircraft panels:

5.1.2.1 Overhead Switch Panel



Figure 27 – Overhead switch panel

Id	Function	Applicable custom events	Parameters	Type
1	Position (nav) lights	POSITION_LIGHTS_TOGGLE	None	Single-shot
		POSITION_LIGHTS_SET	0 = off, 1 = on	Single-shot
2	Instrument (panel) lights	PANEL_LIGHTS_SWITCH_TOGGLE	None	Single-shot
		PANEL_LIGHTS_SWITCH_SET	0 = off, 1 = on	Single-shot
3	Anti-collision (beacon) lights	COLLISION_LIGHTS_TOGGLE	None	Single-shot
		COLLISION_LIGHTS_SET	0 = off, 1 = on	Single-shot
4	Difficulty level setting	DIFFICULTY_LEVEL_ROTARY_SET	1 to 5	Single-shot
		DIFFICULTY_LEVEL_ROTARY_INC	None	Single-shot
		DIFFICULTY_LEVEL_ROTARY_DEC	None	Single-shot
5	Heat/Vent	HEAT_VENT_SWITCH_SET	0 = off, 1 = heat, 2 = vent	Single-shot
		HEAT_VENT_SWITCH_CYCLE	None	Single-shot
6	Generator field CB	GENERATOR_FIELD_CB_TOGGLE	None	Single-shot

		GENERATOR_FIELD_CB_SET	0 = off, 1 = on	Single-shot
7	Generator circuit reset CB	GENERATOR_RESET_CB_TOGGLE	None	Single-shot
		GENERATOR_RESET_CB_SET	0 = off, 1 = on	Single-shot
8	Directional gyro & attitude indicator	DIR_GYRO_SWITCH_TOGGLE	None	Single-shot
		DIR_GYRO_SWITCH_SET	0 = off, 1 = on	Single-shot
9	Pitot heater	PITOT_HEAT_SWITCH_TOGGLE	None	Single-shot
		PITOT_HEAT_SWITCH_SET	0 = off, 1 = on	Single-shot
10	Avionics master	AVIONICS_SWITCH_TOGGLE	None	Single-shot
		AVIONICS_SWITCH_SET	0 = off, 1 = on	Single-shot
11	Battery master	BATTERY_SWITCH_TOGGLE	None	Single-shot
		BATTERY_SWITCH_SET	0 = off, 1 = on	Single-shot
12	Generator master	GENERATOR_SWITCH_TOGGLE	None	Single-shot
		GENERATOR_SWITCH_SET	0 = off, 1 = on	Single-shot
Events for functions that have no 2D panel graphic:				
13	Rotor brake	ROTOR_BRAKE_LEVER_TRIGGER	None	Momentary
		ROTOR_BRAKE_LEVER_SET	0 = off, 1 = on	Single-shot

Table 5- Overhead switch panel custom events

5.1.2.2 Overhead Breaker Panel



Figure 28 – Overhead breaker panel

Note that where equipment is not fitted to a particular aircraft, (such as there being a lack of hoist on all but the Utility model), the relevant circuit breakers are missing, replaced instead with rubber bungs.

Id	Function	Applicable custom events	Parameters	Type
1	Toggle headphones sound set mode	CREW_HEADPHONES_TOGGLE	None	Single-shot
		CREW_HEADPHONES_SET	0 = off, 1 = on	Single-shot
2	Hoist power (controls ability to raise/lower hoist cable)	HOIST_POWER_CB_TOGGLE	None	Single-shot
		HOIST_POWER_CB_SET	0 = off, 1 = on	Single-shot
3	Hoist/sling cable cut power (Non-functional.)	CABLE_CUT_CB_TOGGLE	None	Single-shot
		CABLE_CUT_CB_SET	0 = off, 1 = on	Single-shot
4	Turn/slip indicator power	TURN_IND_CB_TOGGLE	0 = off, 1 = on	Single-shot
		TURN_IND_CB_SET	None	Single-shot
5	DME power (supplies power to DME control on pedestal panel)	DME_CB_TOGGLE	0 = off, 1 = on	Single-shot
		DME_CB_SET	None	Single-shot
6	Transponder power (supplies power to transponder control on pedestal panel)	XPNDR_CB_TOGGLE	None	Single-shot
		XPNDR_CB_SET	0 = off, 1 = on	Single-shot
7	VHF-2 power (supplies power to com/nav 2 radio control on pedestal panel)	VHF2_CB_TOGGLE	None	Single-shot
		VHF2_CB_SET	0 = off, 1 = on	Single-shot
8	GPS power (supplies power to GPS) (Non-functional)	GPS_CB_TOGGLE	None	Single-shot
		GPS_CB_SET	0 = off, 1 = on	Single-shot
9	ADF power (supplies power to adf control on pedestal panel)	ADF_CB_TOGGLE	None	Single-shot
		ADF_CB_SET	0 = off, 1 = on	Single-shot
10	Intercom speaker power (Non-functional)	ICS_SPKR_CB_TOGGLE	None	Single-shot
		ICS_SPKR_CB_SET	0 = off, 1 = on	Single-shot
11	VHF-1 power (supplies power to com/nav 1 radio control on pedestal panel)	VHF1_CB_TOGGLE	None	Single-shot
		VHF1_CB_SET	0 = off, 1 = on	Single-shot
12	Cargo hook power (controls ability to pick-up/release from hoist/sling.)	CARGO_HOOK_CB_TOGGLE	None	Single-shot
		CARGO_HOOK_CB_SET	0 = off, 1 = on	Single-shot
13	Heater control power	HTR_CTRL_CB_TOGGLE	None	Single-shot
		HTR_CTRL_CB_SET	0 = off, 1 = on	Single-shot
14	Heater/Vent power	HTR_PWR_CB_TOGGLE	None	Single-shot
		HTR_PWR_CB_SET	0 = off, 1 = on	Single-shot
15	SAS control power (supplies	SAS_CONT_CB_TOGGLE	None	Single-shot

	power to the SAS actuators, permitting selection of SAS modes of the control on the pedestal panel)	SAS_CONT_CB_SET	0 = off, 1 = on	Single-shot
16	SAS inverter power (supplies power to SAS control on pedestal panel.)	SAS_INVTR_CB_TOGGLE	None	Single-shot
		SAS_INVTR_CB_SET	0 = off, 1 = on	Single-shot
17	Landing light power (supplies power to the landing lights circuit)	LDG_LT_CB_TOGGLE	None	Single-shot
		LDG_LT_CB_SET	0 = off, 1 = on	Single-shot
18	Instrument light power (supplies power to the panel instrument lights circuit)	INST_LT_CB_TOGGLE	None	Single-shot
		INST_LT_CB_SET	0 = off, 1 = on	Single-shot
19	Cockpit light power (supplies power to the cabin spot light circuit)	COCKPIT_LT_CB_TOGGLE	None	Single-shot
		COCKPIT_LT_CB_SET	0 = off, 1 = on	Single-shot
20	Caution (Annunciator) lights power (supplies power to the annunciator lights and audio alarm circuit)	CAUTION_LT_CB_TOGGLE	None	Single-shot
		CAUTION_LT_CB_SET	0 = off, 1 = on	Single-shot
21	Turbine over-temp power (supplies power to the TOT light circuit on the main panel)	TOT_IND_CB_TOGGLE	None	Single-shot
		TOT_IND_CB_SET	0 = off, 1 = on	Single-shot
22	Aft fuel pump power	AFT_FUEL_PUMP_CB_TOGGLE	None	Single-shot
		AFT_FUEL_PUMP_CB_SET	0 = off, 1 = on	Single-shot
23	Forward fuel pump power	FORE_FUEL_PUMP_CB_TOGGLE	None	Single-shot
		FORE_FUEL_PUMP_CB_SET	0 = off, 1 = on	Single-shot
24	Fuel quantity/pressure indicators power (supplies power to both fuel quantity and pressure gauges on main panel and FUEL LOW signal on annunciator.)	FUEL_QTY_CB_TOGGLE	None	Single-shot
		FUEL_QTY_CB_SET	0 = off, 1 = on	Single-shot
26	Fuel valve power (supplies power to the solenoid driving worm-screw fuel valve switch on the main panel)	FUEL_VALVE_CB_TOGGLE	None	Single-shot
		FUEL_VALVE_CB_SET	0 = off, 1 = on	Single-shot
27	Engine and transmission temperature indicators (supplies power to engine and transmission temp gauges on main panel and TRANS OIL TEMP signal on annunciator.)	ENG_XMSN_TEMP_CB_TOGGLE	None	Single-shot
		ENG_XMSN_TEMP_CB_SET	0 = off, 1 = on	Single-shot
28	Hydraulics boost (supplies power to the hydraulics pump	HYD_BOOST_CB_TOGGLE	None	Single-shot

	actuators and hydraulics system switch on the pedestal panel)	HYD_BOOST_CB_SET	0 = off, 1 = on	Single-shot
29	Governor RPM control (supplies power to the governor RPM adjuster beeper switch on the collective pitch lever head.)	GOV_CONT_CB_TOGGLE	None	Single-shot
		GOV_CONT_CB_SET	0 = off, 1 = on	Single-shot
30	Engine ignition power (supplies power to the turbine starter igniters.)	IGN_ENG_CB_TOGGLE	None	Single-shot
		IGN_ENG_CB_SET	0 = off, 1 = on	Single-shot
31	Engine starter power (supplies power to the engine starter circuitry.)	START_ENG_CB_TOGGLE	None	Single-shot
		START_ENG_CB_SET	0 = off, 1 = on	Single-shot
32	Engine anti-ice power (supplies power to the engine anti-icing system and control switch on the pedestal panel.)	ENG_ANTI_ICE_CB_TOGGLE	None	Single-shot
		ENG_ANTI_ICE_CB_SET	0 = off, 1 = on	Single-shot

Table 6- Overheadbreaker panel custom events

5.1.2.3 Pedestal Panel



Figure 29 – Pedestal panel

Id	Function	Applicable custom events	Parameters	Type
1	Enable/disable SAS master mode (defaults to attitude hold)	SAS_MASTER_MODE_BUTTON	None	Single-shot
2	Enable/disable SAS heading hold mode	SAS_HDG_MODE_BUTTON	None	Single-shot
3	Enable/disable SAS navigation hold mode	SAS_NAV_MODE_BUTTON	None	Single-shot
4	Enable/disable SAS speed hold	SAS_SPD_MODE_BUTTON	None	Single-shot

	mode			
5	Enable/disable SAS altitude hold mode	SAS_ALT_MODE_BUTTON	None	Single-shot
6	Enable/disable SAS glide slope hold mode	SAS_VRT_MODE_BUTTON	None	Single-shot
7	Selects com radio 1 for transmission	COM1_TX_SELECT_BUTTON	None	Single-shot
8	Selects com radio 2 for transmission	COM2_TX_SELECT_BUTTON	None	Single-shot
9	Selects both com 1 and 2 receive audio (if disabled then selected radio for transmission is receive)	COM_RECEIVE_ALL_BUTTON	None	Single-shot
10	Enables/disabled nav radio 1 ident audio (See item 10 below)	NAV1_IDENT_SOUND_SWITCH_TOGGLE	None	Single-shot
11	Enables/disabled nav radio 2 ident audio (See item ?? below)	NAV2_IDENT_SOUND_SWITCH_TOGGLE	None	Single-shot
12	Enables/ADF ident audio	ADF_IDENT_SOUND_BUTTON	None	Single-shot
13	Enables/disables DME ident audio	DME_IDENT_SOUND_BUTTON	None	Single-shot
14	Enables/disables marker ident audio	MARKER_IDENT_SOUND_BUTTON	None	Single-shot
15	Enables/disables speaker ident audio (non-functional)	SPEAKER_IDENT_SOUND_BUTTON	None	Single-shot
16	Com/nav 1 radio power switch	COM1_POWER_SWITCH_TOGGLE	None	Single-shot
		COM1_POWER_SWITCH_SET	0 = off, 1 = on	Single-shot
17	Swaps com 1 active and back-up frequencies	COM1_FREQ_SWAP_BUTTON	None	Single-shot
18	Com 1 back-up frequency adjuster (inner and outer rotary knobs)	COM1_INC_WHOLE_TRIGGER	None	Momentary
		COM1_INC_FRACT_TRIGGER	None	Momentary
		COM1_DEC_WHOLE_TRIGGER	None	Momentary
		COM1_DEC_FRACT_TRIGGER	None	Momentary

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19	Nav 1 ident audio switch (tied to button 10)	NAV1_IDENT_SOUND_SWITCH_TOGGLE	None	Single-shot
		NAV1_IDENT_SOUND_SWITCH_SET	0 = off, 1 = on	Single-shot
20	Swaps nav 1 active and back-up frequencies	NAV1_FREQ_SWAP_BUTTON	None	Single-shot
21	Nav 1 back-up frequency adjuster (inner and outer rotary knobs)	NAV1_INC_WHOLE_TRIGGER	None	Momentary
		NAV1_INC_FRACT_TRIGGER	None	Momentary
		NAV1_DEC_WHOLE_TRIGGER	None	Momentary
		NAV1_DEC_FRACT_TRIGGER	None	Momentary
22	Com/nav 2 radio power switch	COM2_POWER_SWITCH_TOGGLE	None	Single-shot
		COM2_POWER_SWITCH_SET	0 = off, 1 = on	Single-shot
23	Swaps com 2 active and back-up frequencies	COM2_FREQ_SWAP_BUTTON	None	Single-shot
24	Com 2 back-up frequency adjuster (inner and outer rotary knobs)	COM2_INC_WHOLE_TRIGGER	None	Momentary
		COM2_INC_FRACT_TRIGGER	None	Momentary
		COM2_DEC_WHOLE_TRIGGER	None	Momentary
		COM2_DEC_FRACT_TRIGGER	None	Momentary
25	Nav 2 ident audio switch (tied to button 10)	NAV2_IDENT_SOUND_SWITCH_TOGGLE	None	Single-shot
		NAV2_IDENT_SOUND_SWITCH_SET	0 = off, 1 = on	Single-shot
26	Swaps nav 2 active and back-up frequencies	NAV2_FREQ_SWAP_BUTTON	None	Single-shot
27	Nav 2 back-up frequency adjuster (inner and outer rotary knobs)	NAV2_INC_WHOLE_TRIGGER	None	Momentary
		NAV2_INC_FRACT_TRIGGER	None	Momentary
		NAV2_DEC_WHOLE_TRIGGER	None	Momentary
		NAV2_DEC_FRACT_TRIGGER	None	Momentary
28	Toggle ANT/ADF mode (main panel ADF instrument disabled in ANT mode)	ADF_ADF_ANT_BUTTON	None	Single-shot
29	Toggle ADF BFO mode (BFO not supported)	ADF_BFO_BUTTON	None	Single-shot
30	Select ADF back-up freq display mode / swap active and backup frequencies	ADF_FREQ_BUTTON	None	Single-shot
31	Select ADF timer display mode / switch between flight and	ADF_FLT_ET_BUTTON	None	Single-shot

	elapsed timers			
32	Set/reset ADF elapsed timer (enters programming mode when held for 4 seconds so set to repeat whilst held)	ADF_SET_RESET_BUTTON_TRIGGER	None	Momentary
33	ADF power switch	ADF_POWER_SWITCH_TOGGLE	None	Single-shot
		ADF_POWER_SWITCH_SET	0 = off, 1 = on	Single-shot
34	ADF frequency / timer adjuster inner and outer rotary and fraction selection mode (When input mode = 1, inner rotary inc/decs ____N, when = 0 incs/decs __NN._) Outer always inc/decs NN__.)	ADF_INPUT_ROTARY_MODE_TOGGLE	None	Single-shot
		ADF_INPUT_ROTARY_MODE_SET	0 = whole, 1 = fraction	Single-shot
		ADF_INC_ROTARY_OUTER_TRIGGER	None	Momentary
		ADF_DEC_ROTARY_OUTER_TRIGGER	None	Momentary
		ADF_INC_ROTARY_INNER_TRIGGER	None	Momentary
		ADF_DEC_ROTARY_INNER_TRIGGER	None	Momentary
35	Set DME mode (Remote Tuning, Freq display, Ground Speed / Time to Station) (Remote Tuning not implemented, operates as per GS/T mode)	DME_MODE_SWITCH_CYCLE	None	Single-shot
		DME_MODE_SWITCH_SET	0 = RMT, 1 = FREQ, 2 = GS/T	Single-shot
36	DME power switch	DME_POWER_SWITCH_TOGGLE	None	Single-shot
		DME_POWER_SWITCH_SET	0 = off, 1 = on	Single-shot
37	DME nav channel input select	DME_CHANNEL_SWITCH_TOGGLE	None	Single-shot
		DME_CHANNEL_SWITCH_SET	1 = nav 1, 2 = nav 2	Single-shot
38	Transponder transmit ident button (non-functional)	XPNDR_IDT_BUTTON	None	Single-shot
39	Transponder digit inputs (if less than 4 digits entered within 5 seconds of the first, entry is cancelled)	XPNDR_0_BUTTON	None	Single-shot
		XPNDR_1_BUTTON	None	Single-shot
		XPNDR_2_BUTTON	None	Single-shot
		XPNDR_3_BUTTON	None	Single-shot
		XPNDR_4_BUTTON	None	Single-shot
		XPNDR_5_BUTTON	None	Single-shot
		XPNDR_6_BUTTON	None	Single-shot

		XPNDR_7_BUTTON	None	Single-shot
40	Transponder clear entry button (cancels input of last digit)	XPNDR_CLR_BUTTON	None	Single-shot
41	Transponder power / mode button (power off only blanks display, underlying sim is always squawking when master avionics power is on, other switch position have no function.)	EVENT_XPNDR_KNOB_CYCLE	None	Single-shot
		XPNDR_KNOB_SET	None	Single-shot
		XPNDR_KNOB_LEFT	None	Single-shot
		XPNDR_KNOB_RIGHT	None	Single-shot
42	Transponder squawk VFR code button	XPNDR_VFR_BUTTON	None	Single-shot
43	Inlet filter bypass switch	INLET_BYPASS_SWITCH_TOGGLE	None	Single-shot
		INLET_BYPASS_SWITCH_SET	0 = off, 1 = on	Single-shot
44	Caution lights (annunciator brightness switch)	CAUTION_LIGHT_SWITCH_TOGGLE	None	Single-shot
		CAUTION_LIGHT_SWITCH_SET	0 = off, 1 = on	Single-shot
45	Engine anti-icing switch	ANTI_ICE_SWITCH_TOGGLE	None	Single-shot
		ANTI_ICE_SWITCH_SET	0 = off, 1 = on	Single-shot
46	Hydraulics boost switch	HYDRAULICS_SWITCH_TOGGLE	None	Single-shot
		HYDRAULICS_SWITCH_SET	0 = off, 1 = on	Single-shot
Events for functions that have no 2D panel graphic:				
47	Manual switch to pause flight timer (if not automatically set to pause on ground in settings.)	ADF_FLT_TIMER_START_STOP_TOGGLE	None	Single-shot
		ADF_FLT_TIMER_START_STOP_SET	0 = off, 1 = on	Single-shot

Table 7- Overhead breaker panel custom events

5.1.2.4 Collective Pitch Lever Panel



Figure 30 – Collective Pitch Lever panel

Id	Function	Applicable custom events	Parameters	Type
1	Landing (and taxi) lights switch	LANDING_LIGHTS_SWITCH_CYCLE	None	Single-shot
		LANDING_LIGHTS_SWITCH_SET	0 = off, 1 = FWD on, 2 = BOTH on	Single-shot
		LANDING_LIGHTS_SWITCH_UP	None	Single-shot
		LANDING_LIGHTS_SWITCH_DOWN	None	Single-shot
2	Rotor RPM governor beeper (set to repeat while held)	GOV_BEEP_UP_SWITCH_TRIGGER	None	Momentary
		GOV_BEEP_DOWN_SWITCH_TRIGGER	None	Momentary
3	Engine starter button	STARTER_BUTTON_TRIGGER	None	Momentary, set to repeat whilst held
		STARTER_BUTTON_SET	0 = off, 1 = on	Momentary
4	Throttle idle release latch	IDLE_RELEASE_LATCH_TRIGGER	None	Momentary, set to repeat whilst held

Table 8- Collective pitch lever head panel custom events

5.1.2.5 Main Panel



Figure 31 – Main panel

Id	Function	Applicable custom events	Parameters	Type
1	Annunciator (caution lights) test button	ANNUNCIATOR_TEST_BUTTON_TRIGGER	None	Momentary, set to repeat whilst held
		ANNUNCIATOR_TEST_BUTTON_SET	0 = off, 1 = on (in)	Momentary
2	Warning horn mute button	WARNING_MUTE_BUTTON_TRIGGER	None	Momentary, set to repeat whilst held
		WARNING_MUTE_BUTTON_SET	0 = off, 1 = on (in)	Momentary
3	Adjust ADF compass card	ADF_CARD_INC_TRIGGER	None	Momentary, set to repeat whilst held
		ADF_CARD_DEC_TRIGGER	None	Momentary, set to repeat whilst held
4	Raise/Lower Attitude Indicator pitch bars and cage	AI_PITCH_BARS_UP_TRIGGER	None	Momentary, set to repeat whilst held
		AI_PITCH_BARS_DOWN_TRIGGER	None	Momentary, set to repeat whilst held

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		AI_CAGE_KNOB_TRIGGER	None	Momentary, set to repeat whilst held
		AI_CAGE_KNOB_SET	0 = Re-cage, 1 = Un-cage	Momentary
5	Altitude indicator Kohlsman settings	EVENT_ALTITUDE_KOHLSMAN_INC_TRIGGER	None	Momentary, set to repeat whilst held
		EVENT_ALTITUDE_KOHLSMAN_DEC_TRIGGER	None	Momentary, set to repeat whilst held
6	GPS drives Nav 1 switch (determines whether GPS is driving HSI gauge and autopilot)	GPS_DRIVES_NAV1_SWITCH_TOGGLE	None	Single-shot
		GPS_DRIVES_NAV1_SWITCH_SET	0 = Nav radio 1 drive HSI, 1 = GPS drives HSI	Single-shot
7	Turbine Over-Temp (TOT) light test	TOT_TEST_BUTTON_TRIGGER	None	Momentary, set to repeat whilst held
		TOT_TEST_BUTTON_SET	0 = off, 1 = on (in)	Momentary
8	HSI (VOR1) Omni-Bearing Selector (OBS) adjust	HSI_VOR1_OBS_INC_TRIGGER	None	Momentary, set to repeat whilst held
		HSI_VOR1_OBS_DEC_TRIGGER	None	Momentary, set to repeat whilst held
9	HSI (VOR1) heading bug adjust	HSI_HDG_BUG_INC_TRIGGER	None	Momentary, set to repeat whilst held
		HSI_HDG_BUG_DEC_TRIGGER	None	Momentary, set to repeat whilst held
10	Clock mode settings	CLOCK_OAT_BUTTON	None	Single-shot
		CLOCK_MODE_BUTTON	None	Single-shot
		CLOCK_STSP_BUTTON	None	Single-shot
11	Radar altimeter Decision Height (DH) audible warning cancel	RAD_ALT_MUTE_ALARM_BUTTON	None	Single-shot
12	Radar altimeter test button	RAD_ALT_TEST_BUTTON_TRIGGER	None	Momentary, set to repeat whilst held
		RAD_ALT_TEST_BUTTON_SET	0 = off, 1 = on (in)	Momentary
13	Radar altimeter altitude bug adjust	RAD_ALT_BUG_INC_TRIGGER	Default none, or multiple 1-10	Momentary, set to repeat whilst held
		RAD_ALT_BUG_DEC_TRIGGER	Default none, or multiple 1-10	Momentary, set to repeat whilst held

14	VOR2 Omni-Bearing Selector (OBS) adjust	VOR2_OBS_INC_TRIGGER	None	Momentary, set to repeat whilst held
		VOR2_OBS_DEC_TRIGGER	None	Momentary, set to repeat whilst held
15	Fuel valve solenoid switch	FUEL_VALVE_SWITCH_TOGGLE	None	Single-shot
		FUEL_VALVE_SWITCH_SET	0 = off, 1 = on	Single-shot

Table 9- Main panel custom events

5.1.2.6 Miscellaneous Non-Panel Events

The following table lists additional custom events which are not associated with any panel switches:

Id	Function	Applicable custom events	Parameters	Type
1	External power connection (on ground only) (Applies 28 Volts DC to main bus, disconnects on lift-off.)	EXTERNAL_POWER_TOGGLE	None	Single-shot
		EXTERNAL_POWER_SET	0 = off, 1 = on	Single-shot
2	Cabin (spot) lights	CABIN_LIGHTS_SWITCH_TOGGLE	None	Single-shot
		CABIN_LIGHTS_SWITCH_SET	0 = off, 1 = on	Single-shot
3	Nudge force-trim or adjust SAS attitude	CYCLIC_TRIM_LEFT_TRIGGER	Default none, or multiplier 1 - 10	Momentary, set to repeat whilst held
		CYCLIC_TRIM_RIGHT_TRIGGER	Default none, or multiplier 1 - 10	Momentary, set to repeat whilst held
		CYCLIC_TRIM_FWD_TRIGGER	Default none, or multiplier 1 - 10	Momentary, set to repeat whilst held
		CYCLIC_TRIM_BACK_TRIGGER	Default none, or multiplier 1 - 10	Momentary, set to repeat whilst held
4	Reset force-trim/SAS (cancels any trim deviation from calibrated cyclic stick centre for either trim or SAS)	CYCLIC_TRIM_SAS_RESET	None	Single-shot
5	Set cyclic trim/SAS attitude (If using trim, to be held while cyclic is relaxed to centre, if SAS pressed and released immediately to set new attitude, press and hold for 1.5 seconds)	CYCLIC_TRIM_SAS_BUTTON_TRIGGER	Momentary	Momentary, set to repeat whilst held
		CYCLIC_TRIM_SAS_BUTTON_SET	None	Momentary

	to enable/disable SAS)			
6	SAS mode disconnect (disconnects advanced autopilot mode if on, disconnect SAS attitude if not.)	SAS_DISCONNECT	None	Single-shot
7	Calibrate cyclic centre (sets cyclic axis deflection zero-point to current axes deflections – for un-sprung cyclic sticks.)	CALIBRATE_CYCLIC_CENTRE	None	Single-shot

Table 10- Miscellaneous custom events

6 GUIDANCE ON LOCATION OF SHARED FILES

The DodoSim 206 V2 is delivered in several different aircraft variants. Many panel graphic, sound and configuration files are common to all variants. Common files are all located within the DodoSim_206_LowSkid variant folder. To ensure all variants are displayed and operate as expected, do not delete or rename the DodoSim_206_LowSkid directory, even if the particular variant is not to be used within your installation.

Should you wish to remove the low skid variant from the simulation's aircraft selection dialog in order to prevent it from being selected for flight, rename the aircraft.cfg file within the DodoSim_206_LowSkid directory to aircraft.cfgBAK.

7 GUIDANCE FOR USING THIRD-PARTY RADIO HARDWARE

The radio-stack equipment modelled in the DodoSim 206 V2 software extends and replaces some default behaviour of the native radio stacks as used in the default aircraft, (specifically the Com/Nav 1 and 2 radios, ADF, DME, Transponder and Audio control panel), which are controlled by some external third-party hardware radio modules. If using external modules, then it is recommended that the relevant systems be disabled in the DodoSim 206 V2 cockpit, permitting native events to pass-through unhindered.

To disable items, open the panel.cfg in the:

{platform root}/SimObjects/Rotorcraft/DodoSim_206_LowSkid/panel

directory, locate the relevant items under the [Window03] and [VCockpit01] sections and comment them out with //.

Relevant equipment names are: CommAudioControl, NavComm1Control, NavComm2Control, ADFControl, DMEControl and XpndrControl.

Note that when radio modules are disabled in the DodoSim 260 panel.cfg, the respective circuit breakers on the overhead breaker panel serve no function.

8 FSUIPC DATA EXPORT

The previous DodoSim 2006 FSX home version software included the ability to export correct gauge values to FSUIPC offset addresses with the aid of additional hardware utility software provided upon request by DodoSim. This is useful for driving SimKits gauges, external or 3D party gauges, etc. in hardware cockpit environments.

This freely available version of the DodoSim 206 V2 includes this feature natively and as such the hardware utility software is not required.

9 APPENDIX 1 – DODOSIM 206 V2.EVT FILE CONTENTS

```
// This is an FSUIPC .evt Simconnect custom event definition file.
// This file is specifically and only usable for the DodoSim 206 V2.
// Manual installation:
// To use, place in{ P3D / FSX } / Modules folder alongside FSUIPC.ini,
// Add entry to FSUIPC.ini thus:
// [EventFiles]
// 0=DodoSim206

// Automatic installation:
// If this file has been generated since FSX/P3D was last started, it should be restarted
// before FSUIPC is able to use this file.
// If this file is new to this directory then FSUIPC will detect it and add it into its .ini
// file the next time it loads.
//
// Events will then be listed in FSUPIC along with native sim events and available for
// button/key binding when FSX/P3D is next started.

// Events ending in _SET require an integer parameter value.
// Unless specified otherwise, use parameter of 1 for on, 0 for off for _SET events.
// Events ending in _TRIGGER require no parameter are momentary buttons/switches. The state
// will revert when the event ceases to be received.
// (Select 'Control to repeat while held' in FSUIPC to cause continued triggering.)
// Events ending in _TOGGLE require no parameter and will alternate state.
// Events that do not end in _SET or _TRIGGER are one-shot actions, requiring no parameter and
// no continued triggering while held.
//
// Note that since FSUIPC sort these entries internally alphabetically and then indexes as
// such.
// Therefore, if this file is updated with new events as part of a future upgrade, the indexing
// inside FSUIPC.ini may become
// out of step with the entries in this file and you may have to reset and re-map your bindings
// in order for them to trigger the correct events.
//
// Exported from DodoSim 206 V2, V2.0a (Build 11-05-19/1)
// Copyright©2019 DodoSim. All Rights Reserved.

[EVENTS]
// Trigger rotor brake (no parameter)
1=DS206.ROTOR_BRAKE_LEVER_TRIGGER
// Set rotor brake % (param value 0 to 100)
2=DS206.ROTOR_BRAKE_LEVER_SET
// Press annunciator test button (no parameter)
3=DS206.ANNUNCIATOR_TEST_BUTTON_TRIGGER
// Set annunciator test button (param value 0 or 1)
4=DS206.ANNUNCIATOR_TEST_BUTTON_SET
// Press warning mute button (no parameter)
5=DS206.WARNING_MUTE_BUTTON_TRIGGER
// Set warning mute button (param value 0 or 1)
6=DS206.WARNING_MUTE_BUTTON_SET
// Trigger attitude indicator cage button (no parameter)
7=DS206.AI_CAGE_KNOB_TRIGGER
// Set attitude indicator cage button (param value 0 or 1)
8=DS206.AI_CAGE_KNOB_SET
// Increment AI pitch bars (no parameter)
9=DS206.AI_PITCH_BARS_UP_TRIGGER
// Decrement AI pitch bars (no parameter)
10=DS206.AI_PITCH_BARS_DOWN_TRIGGER
// Toggle Nav/GPS switch (no parameter)
11=DS206.GPS_DRIVES_NAV1_SWITCH_TOGGLE
// Sets Nav/GPS switch (param value 0 or 1)
12=DS206.GPS_DRIVES_NAV1_SWITCH_SET
// (no parameter)
13=DS206.TOT_TEST_BUTTON_TRIGGER
// Set turbine over temp test button (param value 0 or 1)
14=DS206.TOT_TEST_BUTTON_SET
// Press clock OAT button (no parameter)
15=DS206.CLOCK_OAT_BUTTON
// Press clock MODE button (no parameter)
16=DS206.CLOCK_MODE_BUTTON
// Press clock START/STOP button (no parameter)
17=DS206.CLOCK_STSP_BUTTON
// Press to mute radar altimeter audio (no parameter)
18=DS206.RAD_ALT_MUTE_ALARM_BUTTON
```

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```
// Increment radar altimeter bug, (param is multiplier) (param value 1 to 10)
19=DS206.RAD_ALT_BUG_INC_TRIGGER
// Decrement radar altimeter bug, (param is multiplier) (param value 1 to 10)
20=DS206.RAD_ALT_BUG_DEC_TRIGGER
// Press to test radar altimeter (no parameter)
21=DS206.RAD_ALT_TEST_BUTTON_TRIGGER
// Set radar altimeter test button state (param value 0 or 1)
22=DS206.RAD_ALT_TEST_BUTTON_SET
// Rotate ADF card clockwise (no parameter)
23=DS206.ADF_CARD_INC_TRIGGER
// Rotate ADF card counter-clockwise (no parameter)
24=DS206.ADF_CARD_DEC_TRIGGER
// Set fuel valve switch (param value 0 or 1)
25=DS206.FUEL_VALVE_SWITCH_SET
// Toggle fuel valve switch (no parameter)
26=DS206.FUEL_VALVE_SWITCH_TOGGLE
// Increment altitude indicator Kohlsman setting (no parameter)
27=DS206.ALTITUDE_KOHLSMAN_INC_TRIGGER
// Increment VOR1 (HSI) OBS (no parameter)
28=DS206.HSI_VOR1_OBS_INC_TRIGGER
// Decrement VOR1 (HSI) OBS (no parameter)
29=DS206.HSI_VOR1_OBS_DEC_TRIGGER
// Increment VOR2 OBS (no parameter)
30=DS206.VOR2_OBS_INC_TRIGGER
// Decrement VOR2 OBS (no parameter)
31=DS206.VOR2_OBS_DEC_TRIGGER
// Increment HSI heading bug (no parameter)
32=DS206.HSI_HDG_BUG_INC_TRIGGER
// Decrement HSI heading bug (no parameter)
33=DS206.HSI_HDG_BUG_DEC_TRIGGER
// Toggle crew headphones mode (no parameter)
34=DS206.CREW_HEADPHONES_TOGGLE
// Set crew headphones mode (param value 0 or 1)
35=DS206.CREW_HEADPHONES_SET
// Toggle hoist power breaker (no parameter)
36=DS206.HOIST_POWER_CB_TOGGLE
// Set hoist power breaker (param value 0 or 1)
37=DS206.HOIST_POWER_CB_SET
// Toggle cable cut breaker (no parameter)
38=DS206.CABLE_CUT_CB_TOGGLE
// Set cable cut breaker (param value 0 or 1)
39=DS206.CABLE_CUT_CB_SET
// Toggle turn indicator breaker (no parameter)
40=DS206.TURN_IND_CB_TOGGLE
// Set turn indicator breaker (param value 0 or 1)
41=DS206.TURN_IND_CB_SET
// Toggle DME breaker (no parameter)
42=DS206.DME_CB_TOGGLE
// Set DME breaker (param value 0 or 1)
43=DS206.DME_CB_SET
// Toggle Transponder breaker (no parameter)
44=DS206.XPNDR_CB_TOGGLE
// Set Transponder breaker (param value 0 or 1)
45=DS206.XPNDR_CB_SET
// Toggle VHF 2 radio speaker breaker (no parameter)
46=DS206.VHF2_CB_TOGGLE
// Set VHF 2 radio speaker breaker (param value 0 or 1)
47=DS206.VHF2_CB_SET
// Toggle GPS breaker (no parameter)
48=DS206.GPS_CB_TOGGLE
// Set GPS breaker (param value 0 or 1)
49=DS206.GPS_CB_SET
// Toggle ADF breaker (no parameter)
50=DS206.ADF_CB_TOGGLE
// Set ADF breaker (param value 0 or 1)
51=DS206.ADF_CB_SET
// Toggle intercom speaker breaker (no parameter)
52=DS206.ICS_SPKR_CB_TOGGLE
// Set intercom speaker breaker (param value 0 or 1)
53=DS206.ICS_SPKR_CB_SET
// Toggle VHF 1 radio speaker breaker (no parameter)
54=DS206.VHF1_CB_TOGGLE
// Set VHF 1 radio speaker breaker (param value 0 or 1)
55=DS206.VHF1_CB_SET
// Toggle cargo hook breaker (no parameter)
56=DS206.CARGO_HOOK_CB_TOGGLE
```



```
// Set cargo hook breaker (param value 0 or 1)
57=DS206.CARGO_HOOK_CB_SET
// Toggle heater control breaker (no parameter)
58=DS206.HTR_CTRL_CB_TOGGLE
// Set heater control breaker (param value 0 or 1)
59=DS206.HTR_CTRL_CB_SET
// Toggle heater power breaker (no parameter)
60=DS206.HTR_PWR_CB_TOGGLE
// Set heater power breaker (param value 0 or 1)
61=DS206.HTR_PWR_CB_SET
// Toggle SAS power breaker (no parameter)
62=DS206.SAS_CONT_CB_TOGGLE
// Set SAS power breaker (param value 0 or 1)
63=DS206.SAS_CONT_CB_SET
// Toggle SAS inverter breaker (no parameter)
64=DS206.SAS_INVTR_CB_TOGGLE
// Set SAS inverter breaker (param value 0 or 1)
65=DS206.SAS_INVTR_CB_SET
// Toggle landing light breaker (no parameter)
66=DS206.LDG_LT_CB_TOGGLE
// Set landing light breaker (param value 0 or 1)
67=DS206.LDG_LT_CB_SET
// Toggle instrument light breaker (no parameter)
68=DS206.COCKPIT_LT_CB_TOGGLE
// Set instrument light breaker (param value 0 or 1)
69=DS206.INST_LT_CB_SET
// Toggle cockpit light breaker (no parameter)
70=DS206.COCKPIT_LT_CB_SET
// Set cockpit light breaker (param value 0 or 1)
71=DS206.INST_LT_CB_SET
// Toggle caution light breaker (no parameter)
72=DS206.CAUTION_LT_CB_TOGGLE
// Set caution light breaker (param value 0 or 1)
73=DS206.CAUTION_LT_CB_SET
// Toggle turbine-over-temp light breaker (no parameter)
74=DS206.TOT_IND_CB_TOGGLE
// Set turbine-over-temp light breaker (param value 0 or 1)
75=DS206.TOT_IND_CB_SET
// Toggle aft fuel pump breaker (no parameter)
76=DS206.AFT_FUEL_PUMP_CB_TOGGLE
// Set aft fuel pump breaker (param value 0 or 1)
77=DS206.AFT_FUEL_PUMP_CB_SET
// Toggle fore fuel pump breaker (no parameter)
78=DS206.FORE_FUEL_PUMP_CB_TOGGLE
// Set fore fuel pump breaker (param value 0 or 1)
79=DS206.FORE_FUEL_PUMP_CB_SET
// Toggle fuel quantity gauge breaker (no parameter)
80=DS206.FUEL_QTY_CB_TOGGLE
// Set fuel quantity gauge breaker (param value 0 or 1)
81=DS206.FUEL_QTY_CB_SET
// Toggle fuel valve gauge breaker (no parameter)
82=DS206.FUEL_VALVE_CB_TOGGLE
// Set fuel valve gauge breaker (param value 0 or 1)
83=DS206.FUEL_VALVE_CB_SET
// Toggle engine and transmission temp breaker (no parameter)
84=DS206.ENG_XMSN_TEMP_CB_TOGGLE
// Set engine and transmission temp breaker (param value 0 or 1)
85=DS206.ENG_XMSN_TEMP_CB_SET
// Toggle hydraulics boost breaker (no parameter)
86=DS206.HYD_BOOST_CB_TOGGLE
// Set hydraulics boost breaker (param value 0 or 1)
87=DS206.HYD_BOOST_CB_SET
// Toggle governor control breaker (no parameter)
88=DS206.GOV_CONT_CB_TOGGLE
// Set governor control breaker (param value 0 or 1)
89=DS206.GOV_CONT_CB_SET
// Toggle ignition breaker (no parameter)
90=DS206.IGN_ENG_CB_TOGGLE
// Set ignition breaker (param value 0 or 1)
91=DS206.IGN_ENG_CB_SET
// Toggle starter breaker (no parameter)
92=DS206.START_ENG_CB_TOGGLE
// Set starter breaker (param value 0 or 1)
93=DS206.START_ENG_CB_SET
// Toggle engine anti-icing heater breaker (no parameter)
94=DS206.ENG_ANTI_ICE_CB_TOGGLE
```

```

// Set engine anti-icing heater breaker (param value 0 or 1)
95=DS206.ENG_ANTI_ICE_CB_SET
// Toggle anti-collision lights (no parameter)
96=DS206.COLLISION_LIGHTS_TOGGLE
// Set anti-collision lights (param value 0 or 1)
97=DS206.COLLISION_LIGHTS_SET
// Toggle panel lights (no parameter)
98=DS206.PANEL_LIGHTS_SWITCH_TOGGLE
// Set panel lights (param value 0 or 1)
99=DS206.PANEL_LIGHTS_SWITCH_SET
// Toggle position lights (no parameter)
100=DS206.POSITION_LIGHTS_TOGGLE
// Set navigation (position) lights switch (param value 0 or 1)
101=DS206.POSITION_LIGHTS_SET
// Set difficulty rotary (param value 1 to 5)
102=DS206.DIFFICULTY_LEVEL_ROTARY_SET
// Inc difficulty rotary (no parameter)
103=DS206.DIFFICULTY_LEVEL_ROTARY_INC
// Dec difficulty rotary (no parameter)
104=DS206.DIFFICULTY_LEVEL_ROTARY_DEC
// Set heat/vent switch (param value 0 to 2)
105=DS206.HEAT_VENT_SWITCH_SET
// cycle heat/vent switch up (no parameter)
106=DS206.HEAT_VENT_SWITCH_CYCLE
// Toggle generator field breaker (no parameter)
107=DS206.GENERATOR_FIELD_CB_TOGGLE
// Toggle generator field breaker (param value 0 or 1)
108=DS206.GENERATOR_FIELD_CB_SET
// (no parameter)
109=DS206.GENERATOR_RESET_CB_TOGGLE
// Set generator reset breaker (param value 0 or 1)
110=DS206.GENERATOR_RESET_CB_SET
// Toggle directional gyro switch (no parameter)
111=DS206.DIR_GYRO_SWITCH_TOGGLE
// Set directional gyro switch (param value 0 or 1)
112=DS206.DIR_GYRO_SWITCH_SET
// Toggle pitot heating switch (no parameter)
113=DS206.PITOT_HEAT_SWITCH_TOGGLE
// Set pitot heating switch (param value 0 or 1)
114=DS206.PITOT_HEAT_SWITCH_SET
// Toggle master avionics switch (no parameter)
115=DS206.AVIONICS_SWITCH_TOGGLE
// Set master avionics switch (param value 0 or 1)
116=DS206.AVIONICS_SWITCH_SET
// Toggle master battery switch (no parameter)
117=DS206.BATTERY_SWITCH_TOGGLE
// Set master battery switch (param value 0 or 1)
118=DS206.BATTERY_SWITCH_SET
// Toggle generator switch (no parameter)
119=DS206.GENERATOR_SWITCH_TOGGLE
// Set generator switch (param value 0 or 1)
120=DS206.GENERATOR_SWITCH_SET
// (no parameter)
121=DS206.CABIN_LIGHTS_SWITCH_TOGGLE
// Set cockpit lights switch (param value 0 or 1)
122=DS206.CABIN_LIGHTS_SWITCH_SET
// Toggle engage/disengage SAS master mode (no parameter)
123=DS206.SAS_MASTER_MODE_BUTTON
// Toggle to engage/disengage SAS heading mode (no parameter)
124=DS206.SAS_HDG_MODE_BUTTON
// Toggle to engage/disengage SAS navigation mode (no parameter)
125=DS206.SAS_NAV_MODE_BUTTON
// Toggle to engage/disengage SAS speed mode (no parameter)
126=DS206.SAS_SPD_MODE_BUTTON
// Toggle to engage/disengage SAS altitude mode (no parameter)
127=DS206.SAS_ALT_MODE_BUTTON
// Toggle to engage/disengage SAS vertical approach mode (no parameter)
128=DS206.SAS_VRT_MODE_BUTTON
// Select Com 1 for transmit (no parameter)
129=DS206.COM1_TX_SELECT_BUTTON
// Select Com 2 for transmit (no parameter)
130=DS206.COM2_TX_SELECT_BUTTON
// Select both Com 1 and Com 2 for receive (no parameter)
131=DS206.COM_RECEIVE_ALL_BUTTON
// Set nav 1 ident sound switch (param value 0 or 1)
132=DS206.NAV1_IDENT_SOUND_SWITCH_SET

```

```
// Toggle nav 1 sound switch (no parameter)
133=DS206.NAV1_IDENT_SOUND_SWITCH_TOGGLE
// Set nav 2 ident sound switch (param value 0 or 1)
134=DS206.NAV2_IDENT_SOUND_SWITCH_SET
// Toggle nav 2 sound switch (no parameter)
135=DS206.NAV2_IDENT_SOUND_SWITCH_TOGGLE
// Toggle ADF Id audio (no parameter)
136=DS206.ADF_IDENT_SOUND_BUTTON
// Toggle marker ident audio (no parameter)
137=DS206.MARKER_IDENT_SOUND_BUTTON
// Toggle speaker ident audio (no parameter)
138=DS206.SPEAKER_IDENT_SOUND_BUTTON
// Toggle DME Id audio (no parameter)
139=DS206.DME_IDENT_SOUND_BUTTON
// Swap com 1 active/standby frequency (no parameter)
140=DS206.COM1_FREQ_SWAP_BUTTON
// Swap nav 1 active/standby frequency (no parameter)
141=DS206.NAV1_FREQ_SWAP_BUTTON
// Swap com 2 active/standby frequency (no parameter)
142=DS206.COM2_FREQ_SWAP_BUTTON
// Swap nav 2 active/standby frequency (no parameter)
143=DS206.NAV2_FREQ_SWAP_BUTTON
// Com 1 frequency increment whole (no parameter)
144=DS206.COM1_INC_WHOLE_TRIGGER
// Com 1 frequency increment fraction (no parameter)
145=DS206.COM1_INC_FRACT_TRIGGER
// Com 1 frequency decrement whole (no parameter)
146=DS206.COM1_DEC_WHOLE_TRIGGER
// Com 1 frequency decrement fraction (no parameter)
147=DS206.COM1_DEC_FRACT_TRIGGER
// Nav 1 frequency increment whole (no parameter)
148=DS206.NAV1_INC_WHOLE_TRIGGER
// Nav 1 frequency increment fraction (no parameter)
149=DS206.NAV1_INC_FRACT_TRIGGER
// Nav 1 frequency decrement whole (no parameter)
150=DS206.NAV1_DEC_WHOLE_TRIGGER
// Nav 1 frequency decrement fraction (no parameter)
151=DS206.NAV1_DEC_FRACT_TRIGGER
// Com 2 frequency increment whole (no parameter)
152=DS206.COM2_INC_WHOLE_TRIGGER
// Com 2 frequency increment fraction (no parameter)
153=DS206.COM2_INC_FRACT_TRIGGER
// Com 2 frequency decrement whole (no parameter)
154=DS206.COM2_DEC_WHOLE_TRIGGER
// Com 2 frequency decrement fraction (no parameter)
155=DS206.COM2_DEC_FRACT_TRIGGER
// Nav 2 frequency increment whole (no parameter)
156=DS206.NAV2_INC_WHOLE_TRIGGER
// Nav 2 frequency increment fraction (no parameter)
157=DS206.NAV2_INC_FRACT_TRIGGER
// Nav 2 frequency decrement whole (no parameter)
158=DS206.NAV2_DEC_WHOLE_TRIGGER
// Nav 2 frequency decrement fraction (no parameter)
159=DS206.NAV2_DEC_FRACT_TRIGGER
// Com 1 frequency decrement with carry (no parameter)
160=DS206.COM1_DEC_FRACT_CARRY_TRIGGER
// Com 1 frequency increment with carry (no parameter)
161=DS206.COM1_INC_FRACT_CARRY_TRIGGER
// Toggle com 1 power switch (no parameter)
162=DS206.COM1_POWER_SWITCH_TOGGLE
// Set com 1 power switch (param value 0 or 1)
163=DS206.COM1_POWER_SWITCH_SET
// Com 2 frequency decrement with carry (no parameter)
164=DS206.COM2_DEC_FRACT_CARRY_TRIGGER
// Com 2 frequency increment with carry (no parameter)
165=DS206.COM2_INC_FRACT_CARRY_TRIGGER
// Toggle com 2 power switch (no parameter)
166=DS206.COM2_POWER_SWITCH_TOGGLE
// Set com 2 power switch (param value 0 or 1)
167=DS206.COM2_POWER_SWITCH_SET
// Nav 1 frequency decrement with carry (no parameter)
168=DS206.NAV1_DEC_FRACT_CARRY_TRIGGER
// Nav 1 frequency increment with carry (no parameter)
169=DS206.NAV1_INC_FRACT_CARRY_TRIGGER
// Nav 2 frequency decrement with carry (no parameter)
170=DS206.NAV2_DEC_FRACT_CARRY_TRIGGER
```

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```
// Nav 2 frequency increment with carry (no parameter)
171=DS206.NAV2_INC_FRACT_CARRY_TRIGGER
// Start/stop ADF flight timer (no cockpit switch) (param value 0 or 1)
172=DS206.ADF_FLT_TIMER_START_STOP_SET
// Toggle ADF flight timer (no cockpit switch) (no parameter)
173=DS206.ADF_FLT_TIMER_START_STOP_TOGGLE
// Press ADF ADF/ANT button (no parameter)
174=DS206.ADF_ADF_ANT_BUTTON
// Press ADF BFO button (no parameter)
175=DS206.ADF_BFO_BUTTON
// Press ADF FREQ button (no parameter)
176=DS206.ADF_FREQ_BUTTON
// Press ADF FLT/ET button (no parameter)
177=DS206.ADF_FLT_ET_BUTTON
// Press/hold ADF SET/RESET button (no parameter)
178=DS206.ADF_SET_RESET_BUTTON_SET_TRIGGER
// Toggle ADF power switch (no parameter)
179=DS206.ADF_POWER_SWITCH_TOGGLE
// Set ADF power switch (param value 0 or 1)
180=DS206.ADF_POWER_SWITCH_SET
// Set ADF rotary decimal input mode (param value 0 or 1)
181=DS206.ADF_INPUT_ROTARY_MODE_SET
// Toggle ADF rotary decimal input mode (no parameter)
182=DS206.ADF_INPUT_ROTARY_MODE_TOGGLE
// Increment ADF rotary inner (no parameter)
183=DS206.ADF_INC_ROTARY_INNER_TRIGGER
// Decrement ADF rotary inner (no parameter)
184=DS206.ADF_DEC_ROTARY_INNER_TRIGGER
// Increment ADF rotary outer (no parameter)
185=DS206.ADF_INC_ROTARY_OUTER_TRIGGER
// Decrement ADF rotary outer (no parameter)
186=DS206.ADF_DEC_ROTARY_OUTER_TRIGGER
// Toggle DME power switch (no parameter)
187=DS206.DME_POWER_SWITCH_TOGGLE
// Set DME power switch (param value 0 or 1)
188=DS206.DME_POWER_SWITCH_SET
// Toggle DME nav radio input channel switch (no parameter)
189=DS206.DME_CHANNEL_SWITCH_SET
// Set DME nav radio input channel switch (param value 1 or 2)
190=DS206.DME_CHANNEL_SWITCH_TOGGLE
// Set DME mode switch (param value 0 to 2)
191=DS206.DME_MODE_SWITCH_SET
// Cycle DME mode switch (no parameter)
192=DS206.DME_MODE_SWITCH_CYCLE
// Turn Xpndr knob anti-clockwise (no parameter)
193=DS206.XPNDR_KNOB_LEFT
// Turn Xpndr knob clockwise (no parameter)
194=DS206.XPNDR_KNOB_RIGHT
// Set Xpndr knob position (param value 0 to 4)
195=DS206.XPNDR_KNOB_SET
// Cycle Xpndr knob position (no parameter)
196=DS206.XPNDR_KNOB_CYCLE
// Press Xpndr IDT button (no parameter)
197=DS206.XPNDR_IDT_BUTTON
// Press Xpndr 0 button (no parameter)
198=DS206.XPNDR_0_BUTTON
// Press Xpndr 1 button (no parameter)
199=DS206.XPNDR_1_BUTTON
// Press Xpndr 2 button (no parameter)
200=DS206.XPNDR_2_BUTTON
// Press Xpndr 3 button (no parameter)
201=DS206.XPNDR_3_BUTTON
// Press Xpndr 4 button (no parameter)
202=DS206.XPNDR_4_BUTTON
// Press Xpndr 5 button (no parameter)
203=DS206.XPNDR_5_BUTTON
// Press Xpndr 6 button (no parameter)
204=DS206.XPNDR_6_BUTTON
// Press Xpndr CLR button (no parameter)
205=DS206.XPNDR_CLR_BUTTON
// Press Xpndr VFR button (no parameter)
206=DS206.XPNDR_VFR_BUTTON
// Toggle inlet bypass switch (no parameter)
207=DS206.INLET_BYPASS_SWITCH_TOGGLE
// Set inlet bypass switch (param value 0 or 1)
208=DS206.INLET_BYPASS_SWITCH_SET
```

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```
// Toggle caution light brightness switch (no parameter)
209=DS206.CAUTION_LIGHT_SWITCH_TOGGLE
// Set caution light brightness switch (param value 0 or 1)
210=DS206.CAUTION_LIGHT_SWITCH_SET
// Toggle anti-icing switch (no parameter)
211=DS206.ANTI_ICE_SWITCH_TOGGLE
// Set anti-icing switch (param value 0 or 1)
212=DS206.ANTI_ICE_SWITCH_SET
// Toggle hydraulics switch (no parameter)
213=DS206.HYDRAULICS_SWITCH_TOGGLE
// Set hydraulics switch (param value 0 or 1)
214=DS206.HYDRAULICS_SWITCH_SET
// Set force trim button/set new SAS attitude (param value 0 or 1)
215=DS206.CYCLIC_TRIM_SAS_BUTTON_SET
// Press force trim button/set new SAS attitude (no parameter)
216=DS206.CYCLIC_TRIM_SAS_BUTTON_TRIGGER
// Nudge force trim left (param value 0 to 10)
217=DS206.CYCLIC_TRIM_LEFT_TRIGGER
// Nudge force trim right (param value 0 to 10)
218=DS206.CYCLIC_TRIM_RIGHT_TRIGGER
// Nudge force trim forward (param value 0 to 10)
219=DS206.CYCLIC_TRIM_FWD_TRIGGER
// Nudge force trim aftward (param value 0 to 10)
220=DS206.CYCLIC_TRIM_BACK_TRIGGER
// Trigger reset force trim deflection (no parameter)
221=DS206.CYCLIC_TRIM_SAS_RESET
// Press to disconnect SAS (no parameter)
222=DS206.SAS_DISCONNECT
// Calibrate cyclic centre position (no parameter)
223=DS206.CALIBRATE_CYCLIC_CENTRE
// Cycle landing and taxi light switch (no parameter)
224=DS206.LANDING_LIGHTS_SWITCH_CYCLE
// Nudge landing and taxi light switch up (no parameter)
225=DS206.LANDING_LIGHTS_SWITCH_UP
// Nudge landing and taxi light switch down (no parameter)
226=DS206.LANDING_LIGHTS_SWITCH_DOWN
// Set landing light switch (param value 0 to 2)
227=DS206.LANDING_LIGHT_SWITCH_SET
// Trigger idle release latch (no parameter)
228=DS206.IDLE_RELEASE_LATCH_TRIGGER
// Press starter button (no parameter)
229=DS206.STARTER_BUTTON_TRIGGER
// Set starter button (param value 0 or 1)
230=DS206.STARTER_BUTTON_SET
// Inc engine governor (no parameter)
231=DS206.GOV_BEEP_UP_SWITCH_TRIGGER
// Dec engine governor (no parameter)
232=DS206.GOV_BEEP_DOWN_SWITCH_TRIGGER
// Toggle external power (no parameter)
233=DS206.EXTERNAL_POWER_TOGGLE
// Set external power (param value 0 or 1)
234=DS206.EXTERNAL_POWER_SET
```

-End-